Factor-Adjusted Smoothed Rank Estimation for High-Dimensional Accelerated Failure Time Models

Weineng ZHU · Yang JU

Abstract Factor adjustment separates common and feature-specific variation in high-dimensional accelerated failure time (AFT) regression. Under right censoring, the estimated factor coordinates must be connected to the original sparse slope. We develop a smoothed rank estimator with unpenalized clinical and factor effects and an ℓ_1 penalty on molecular coefficients. An exact reference representation preserves the ordinary-model predictor and separates design perturbation from additional-factor approximation. Under common-error, conditional-censoring, local-information, and factor-design conditions, finite-sample coefficient bounds require no moments of the log-time error. The analysis accommodates factors estimated from the same covariate sample. Column-scale control of pairwise fluctuations avoids an additional logarithmic loss. A full-coordinate stationarity criterion transfers the bounds to numerical candidates. In three fixed strong-dependence samples, factor adjustment has lower maximum active curvature and uses fewer gradient evaluations to reach common accuracy. One active-face Newton step completes each unadjusted fit. Training selection yields twelve fits at the prescribed tolerance and distinguishes sparse residual effects from dense factor-mediated prediction.

Keywords Accelerated failure time · Right censoring · Factor adjustment · High-dimensional estimation · Smoothed rank regression · Computational accuracy

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1 Introduction

High-dimensional survival analysis seeks interpretable associations between incomplete event times and many dependent covariates. Molecular measurements often combine a low-dimensional common component with feature-specific variation. In an accelerated failure time (AFT) model, the scientific target is a sparse slope on the observed covariates. Factor adjustment offers a representation of their dependence, but the factors used in estimation are themselves learned from the sample. A statistical analysis must therefore connect the original slope to the estimated factor coordinates while accounting for right censoring. We develop that connection for a smoothed rank estimator.

Rank-based AFT regression estimates conditional log-time associations with an unspecified error distribution (Wei, 1992; Jin et al., 2003). Regularized Gehan estimation, joint clinical adjustment, and scalable high-dimensional algorithms are developed by Cai et al. (2009), Johnson (2009), and Suder and Molstad (2022). Distribution-function and parameter-perturbation smoothing produce differentiable censored-rank criteria (Heller, 2007; Johnson and Strawderman, 2009). Related computational and robust formulations include Choi and Choi (2021), Li et al. (2021), and Seon et al. (2026, preprint, arXiv:2606.31094). Our estimator uses a common scalar Gaussian convolution of the positive-part loss, so the bandwidth enters the analysis through an explicit smoothing-bias term.

Factor adjustment retains the common component as an unpenalized regression block. FarmSelect develops this construction for regularized estimation (Fan et al., 2020); Bayle and Fan (2022, preprint, arXiv:2210.01067) study factor-augmented Cox regression. For censored rank estimation, the estimated factor representation and the patient pairs in the loss share the same covariate sample. This dependence requires a joint treatment of score centering, pairwise fluctuations, smoothing bias, and estimated-design curvature. The unpenalized clinical and factor coefficients must also be included in the error geometry.

Our primary target is $\boldsymbol{\beta}_0$ in the ordinary observed-covariate model

$$\begin{aligned} \boldsymbol{X}_i &= \boldsymbol{B}\boldsymbol{f}_i + \boldsymbol{u}_i, \\ \log T_i &= \boldsymbol{Z}_i^\top \boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i, \end{aligned} \tag{1.1}$$

where $\boldsymbol{X}_i \in \mathbb{R}^p$, $\boldsymbol{Z}_i \in \mathbb{R}^d$, and the factor dimension K is small. The predictor is equivalently $\boldsymbol{Z}_i^\top \boldsymbol{\theta}_0 + \boldsymbol{u}_i^\top \boldsymbol{\beta}_0 + \boldsymbol{f}_i^\top \boldsymbol{\gamma}_0$, with $\boldsymbol{\gamma}_0 = \boldsymbol{B}^\top \boldsymbol{\beta}_0$. We estimate the common and residual components by principal component analysis (PCA), then minimize a smoothed Gehan loss with an ℓ_1 penalty on $\boldsymbol{\beta}$. Clinical and factor coefficients are estimated jointly and remain unpenalized. The estimator allows the factor coefficient to vary freely; it contains the ordinary fitted predictor as a special case while retaining $\boldsymbol{\beta}_0$ as its sparse target. Section 2 also considers an additional factor effect relative to a specified latent representation.

The main contribution is a finite-sample analysis with same-sample factor estimation. An exact reference representation preserves the centered ordinary-model predictor and separates estimated-design perturbation from the approximation associated with an additional factor effect. Conditional on the full design, an event-weighted residual-difference identity centers the unsmoothed score. Pairing and contraction bound the empirical contrasts at the design's column root-mean-square scale, avoiding the extra logarithmic loss of an entrywise-envelope argument. Local event information and a regularization argument that includes the unpenalized block yield coefficient bounds and finite attainment on the same statistical event. Primitive factor conditions supply the required estimated-design geometry. The analysis uses integrable loss differences and accommodates log-time errors without moments under the stated density and censoring conditions.

A second contribution connects statistical and numerical accuracy. A complete stationarity residual transfers the coefficient guarantee to finite numerical candidates on the same fixed-tuning event. Event-row blocking evaluates the full pairwise gradient efficiently. We provide a reference proximal-gradient convergence result and implement the same objective using accelerated iterations with qualified active-face Newton refinement. These components separate the statistical target, the search algorithm, and the accuracy required of its output.

The numerical illustrations connect this framework to computational geometry and training selection. In three fixed strong-dependence samples, factor adjustment has lower maximum active curvature and reaches common stationarity thresholds with fewer gradient evaluations. Residual projections identify weak-curvature directions in the late iterates; one Newton step completes each of the three unadjusted fits on its existing active face. A shared fold-specific selection procedure produces twelve full-training refits at the prescribed tolerance. The selected models also distinguish sparse residual effects from the dense molecular slope induced by the common factors. Each illustration addresses the numerical or representational role of the estimator; independent repeated-sample risk and biomedical applications are separate evaluation targets.

Section 2 defines the observation scheme and regression targets. Section 3 constructs the estimator and frozen prediction map. Sections 4 and 5 develop the statistical and numerical results. Section 6 presents the computational and training-selection illustrations, and Section 7 discusses their implications. Online Resource 1 contains the proofs in Supplementary Appendices A–E and the reproducibility and implementation details in Appendices F–G.

2 Model and estimation targets

2.1 Observed data

Bold letters denote vectors and matrices; indexed scalar entries are italic. For independent patients, we observe

$$\begin{aligned} O_i &= (Y_i, \Delta_i, \boldsymbol{X}_i, \boldsymbol{Z}_i), \\ Y_i &= \log\{\min(T_i, C_i)\}, \\ \Delta_i &= \mathbf{1}\{T_i \le C_i\}, \\ i &= 1, \dots, n, \end{aligned} \tag{2.1}$$

where $T_i \in (0, \infty)$ is the event time, $C_i \in (0, \infty]$ is the censoring time, $\boldsymbol{X}_i \in \mathbb{R}^p$ contains baseline molecular measurements, and $\boldsymbol{Z}_i \in \mathbb{R}^d$ contains prespecified clinical covariates. We consider a single right-censored endpoint observed from time zero, with fully observed baseline covariates and Y_i on the natural-logarithmic scale.

The covariates have prespecified coordinate scales and finite second moments. We absorb population means into the unspecified error location and take $E(\boldsymbol{Z}_i) = E(\boldsymbol{X}_i) = 0$ for notation. The clinical design excludes a constant column; Section 2.6 defines sample centering. The dimension $p = p_n$ and sparsity may grow with n, whereas d and the true factor dimension K are fixed in the theory. Dependence of the joint law on n is suppressed.

2.2 Factor representation and coordinate normalization

Suppose the molecular covariates admit the representation

$$\begin{aligned} \boldsymbol{X}_i &= \boldsymbol{B}\boldsymbol{f}_i + \boldsymbol{u}_i, \\ \boldsymbol{B} \in \mathbb{R}^{p\times K}, &\quad \boldsymbol{f}_i \in \mathbb{R}^K, \quad \boldsymbol{u}_i \in \mathbb{R}^p, \end{aligned} \tag{2.2}$$

with

$$E(\boldsymbol{f}_i) = 0, \quad E(\boldsymbol{u}_i) = 0, \quad E(\boldsymbol{f}_i\boldsymbol{f}_i^\top) = \boldsymbol{I}_K, \quad E(\boldsymbol{f}_i\boldsymbol{u}_i^\top) = 0. \tag{2.3}$$

The factors and idiosyncratic components are latent. Define the full design vector

$$\boldsymbol{D}_i = (\boldsymbol{Z}_i^\top, \boldsymbol{f}_i^\top, \boldsymbol{u}_i^\top)^\top \in \mathbb{R}^P, \qquad q = d + K, \qquad P = p + q.$$

Equation (2.3) imposes second-moment orthogonality between factors and idiosyncratic components, while allowing dependence within and between covariate blocks. Clinical covariates may be associated with either component.

For the pervasive-factor regime, the basic structural conditions are

$$\begin{aligned} c_B\boldsymbol{I}_K \preceq p^{-1}\boldsymbol{B}^\top\boldsymbol{B} \preceq C_B\boldsymbol{I}_K, \\ \|\boldsymbol{\Sigma}_u\|_{\mathrm{op}} \le C_u, \\ \boldsymbol{\Sigma}_u = \mathrm{Cov}(\boldsymbol{u}_i), \end{aligned} \tag{2.4}$$

where the positive constants are uniform in n, p. Equation (2.4) supplies the spectral separation used in factor-adjusted regression (Fan et al., 2020); Section 4 adds distributional, loading, and sparse-design conditions for estimated-factor guarantees.

For an orthogonal $K \times K$ matrix $\boldsymbol{Q}$, the transformation

$$\boldsymbol{f}_i^* = \boldsymbol{Q}\boldsymbol{f}_i, \qquad \boldsymbol{B}^* = \boldsymbol{B}\boldsymbol{Q}^\top \tag{2.5}$$

preserves the common component and (2.3); the factor coefficients transform with the same rotation. Population rotation and sample normalization $\widehat{\boldsymbol{F}}^\top\widehat{\boldsymbol{F}}/n = \boldsymbol{I}_K$ play separate roles. Finite-sample alignment matrices can be nonorthogonal (Bai and Ng, 2013). Section 2.5 establishes slope identification for these coordinate representations.

2.3 AFT models and the sparse coefficient target

The primary model is an ordinary sparse AFT model,

$$\mathcal{M}_0 : \qquad \log T_i = \boldsymbol{Z}_i^\top\boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top\boldsymbol{\beta}_0 + \varepsilon_i. \tag{2.6}$$

Factor adjustment retains $\boldsymbol{\beta}_0$ as the observed-covariate slope and represents dependence through common and residual components. We also consider the extension

$$\begin{aligned} \mathcal{M}_1 : \qquad \log T_i &= \eta_i^0 + \varepsilon_i, \\ \eta_i^0 &= \boldsymbol{Z}_i^\top\boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top\boldsymbol{\beta}_0 + \boldsymbol{f}_i^\top\boldsymbol{\xi}_0 \\ &= \boldsymbol{Z}_i^\top\boldsymbol{\theta}_0 + \boldsymbol{u}_i^\top\boldsymbol{\beta}_0 + \boldsymbol{f}_i^\top\boldsymbol{\gamma}_0, \\ \boldsymbol{\gamma}_0 &= \boldsymbol{B}^\top\boldsymbol{\beta}_0 + \boldsymbol{\xi}_0. \end{aligned} \tag{2.7}$$

Thus $\mathcal{M}_0$ is the submodel $\boldsymbol{\xi}_0 = 0$. Additional factor contributions also arise in factor-augmented hazard regression (Bayle and Fan, 2022, preprint, arXiv:2210.01067). In $\mathcal{M}_1$, the coefficient target is defined relative to the specified latent factor representation.

Write

$$\begin{aligned} \boldsymbol{a}_0 &= (\boldsymbol{\theta}_0^\top, \boldsymbol{\gamma}_0^\top)^\top, \\ \boldsymbol{\vartheta}_0 &= (\boldsymbol{a}_0^\top, \boldsymbol{\beta}_0^\top)^\top, \\ S &= \operatorname{supp}(\boldsymbol{\beta}_0), \\ s &= |S|, \\ k &= s + q. \end{aligned} \tag{2.8}$$

Sparsity is imposed on $\boldsymbol{\beta}_0$; the q clinical and factor coefficients are unpenalized. The target $\boldsymbol{\beta}_0$, its support, and η_i^0 are invariant under (2.5) when $\boldsymbol{\gamma}_0^* = \boldsymbol{Q}\boldsymbol{\gamma}_0$ and $\boldsymbol{\xi}_0^* = \boldsymbol{Q}\boldsymbol{\xi}_0$. Individual factor coordinates transform with the basis. Even for fixed K, the magnitude of $\boldsymbol{\gamma}_0 = \boldsymbol{B}^\top\boldsymbol{\beta}_0$ in $\mathcal{M}_0$ can grow with the sparse signal.

The error is independent of $\boldsymbol{D}_i$ and has a common distribution function F_ε, defining a location-shift model with unrestricted location and no mean or variance requirement. If $F_\varepsilon^{-1}(\tau)$ denotes its generalized quantile, the model gives

$$Q_{T|\boldsymbol{D}_i}(\tau) = \exp\{\eta_i^0 + F_\varepsilon^{-1}(\tau)\}, \qquad 0 < \tau < 1. \tag{2.9}$$

Within covariate support, a unit change in u_{ij} at fixed remaining coordinates multiplies conditional event-time quantiles by $\exp(\beta_{0j})$. The coefficients describe conditional log-time associations: larger η_i^0 corresponds to longer event times. The intercept is absorbed into F_ε and cancels from pairwise residual comparisons.

2.4 Censoring and observable local event information

Assumption 2.1 (Common error and conditional censoring)**.** The triples $(\boldsymbol{D}_i, \varepsilon_i, C_i)$ are independent and identically distributed, and

$$\varepsilon_i \perp \boldsymbol{D}_i, \qquad \varepsilon_i \perp C_i \mid \boldsymbol{D}_i. \tag{2.10}$$

The common error distribution is absolutely continuous, with density f_ε satisfying $f_\varepsilon(t) \le L_\varepsilon < \infty$ for almost every $t \in \mathbb{R}$.

Assumption 2.1 yields $\Pr(\varepsilon_i \le t \mid \boldsymbol{D}_i, C_i) = F_\varepsilon(t)$ and $T_i \perp C_i \mid \boldsymbol{D}_i$. Censoring may depend on any covariate block and may include atoms, such as an administrative end of follow-up. The density condition applies to the event error. Recorded-time rounding is handled by a separately specified observation or tie convention.

In $\mathcal{M}_0$, η_i^0 is measurable with respect to $(\boldsymbol{X}_i, \boldsymbol{Z}_i)$, and Assumption 2.1 also gives $T_i \perp C_i \mid (\boldsymbol{X}_i, \boldsymbol{Z}_i)$ because ε_i is independent of $(\boldsymbol{D}_i, C_i)$. For $\mathcal{M}_1$, conditioning on the full design retains the association that can arise between the additional term $\boldsymbol{f}_i^\top \boldsymbol{\xi}_0$ and censoring after the factors are integrated out.

Define the residual-scale quantities

$$\begin{aligned} C_i^* &= \log C_i - \eta_i^0, \\ R_i &= Y_i - \eta_i^0 = \min(\varepsilon_i, C_i^*), \\ G(t \mid \boldsymbol{D}_i) &= \Pr(C_i^* \ge t \mid \boldsymbol{D}_i). \end{aligned} \tag{2.11}$$

Let $S_\varepsilon(t) = \Pr(\varepsilon_i \ge t)$.

Assumption 2.2 (A common local information interval)**.** There exist $t_\star \in \mathbb{R}$, $A > 0$, $m_\varepsilon > 0$, and $g_0 > 0$ such that, on $I = [t_\star - A, t_\star + A]$,

$$f_\varepsilon(t) \geq m_\varepsilon \quad \text{a.e.}, \qquad \inf_{t \in I} G(t \mid \boldsymbol{D}_i) \geq g_0 \quad \text{a.s.} \tag{2.12}$$

The constants used for uniform bounds are independent of n and p.

Assumption 2.2 supplies a common residual-time interval with positive event density and follow-up probability uniformly over the design, while allowing arbitrary error location. This design-uniform overlap is stronger than control of the marginal censoring proportion. For example, fixed administrative follow-up can violate the overlap condition when η_i^0 is unbounded above.

Conditioning on $D_n = \sigma(\boldsymbol{D}_1, \ldots, \boldsymbol{D}_n)$ fixes factor estimates obtained from the covariates. Patient outcomes remain conditionally independent with potentially different censoring laws. Section 4 uses this conditional structure and treats patients as the independent sampling units.

2.5 Identification of the slope target

Slope uniqueness is governed by the following sparse-design separation condition. It complements the coordinate normalization in Section 2.2.

Assumption 2.3 (No sparse affine alias)**.** For some $\kappa_{\mathrm{pop}} > 0$,

$$\mathrm{Var}(\boldsymbol{D}_i^\top \boldsymbol{v}) \geq \kappa_{\mathrm{pop}} \|\boldsymbol{v}\|_2^2 \quad \text{whenever} \quad \boldsymbol{v} = (\boldsymbol{v}_a^\top, \boldsymbol{v}_\beta^\top)^\top, \quad \|\boldsymbol{v}_\beta\|_0 \leq 2s. \tag{2.13}$$

The unrestricted block $\boldsymbol{v}_a$ allows joint variation in clinical and factor effects. Assumption 2.3 separates a sparse molecular signal from these effects; a positive lower eigenvalue of $\mathrm{Cov}(\boldsymbol{D}_i)$ is sufficient. For $\mathcal{M}_0$, the analogous restriction on $(\boldsymbol{Z}_i, \boldsymbol{X}_i)$ suffices. This condition identifies the population target, while Section 4 introduces the estimated-design cone condition for penalized estimation.

To state the target precisely without assuming integrable log-time errors, consider two independent patients, indexed by 1 and 2. Put

$$e_i^o(\boldsymbol{\vartheta}) = Y_i - \boldsymbol{D}_i^\top \boldsymbol{\vartheta}, \qquad \Theta_s = \{\boldsymbol{\vartheta} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top : \|\boldsymbol{\beta}\|_0 \leq s\}.$$

The superscript o denotes use of the latent design. Define the expected Gehan *contrast*

$$\mathcal{G}(\boldsymbol{\vartheta}, \boldsymbol{\vartheta}_0) = E\Big[\Delta_1 \Big\{ \big(e_2^o(\boldsymbol{\vartheta}) - e_1^o(\boldsymbol{\vartheta})\big)_+ - \big(e_2^o(\boldsymbol{\vartheta}_0) - e_1^o(\boldsymbol{\vartheta}_0)\big)_+ \Big\}\Big]. \tag{2.14}$$

The subtraction is inside the expectation. The 1-Lipschitz positive-part function bounds the integrand by $|(\boldsymbol{D}_1 - \boldsymbol{D}_2)^\top(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0)|$, which is integrable under the covariate moment condition. Thus the Gehan contrast is well defined without an integrable log-time error (Cai et al., 2009; Suder and Molstad, 2022).

Proposition 2.1 (Separation of the rank target)**.** *Under Assumptions 2.1–2.3, let*

$$c_{\mathrm{loc}} = A m_\varepsilon^2 g_0^2, \qquad \rho_A(t) = \begin{cases} t^2, & |t| \le A, \\ 2A|t| - A^2, & |t| > A. \end{cases}$$

For every $\boldsymbol{\vartheta} \in \Theta_s$, with $\boldsymbol{v} = \boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0$,

$$\mathcal{G}(\boldsymbol{\vartheta}, \boldsymbol{\vartheta}_0) \ge \frac{c_{\mathrm{loc}}}{2} E\big[\rho_A\{(\boldsymbol{D}_1 - \boldsymbol{D}_2)^\top \boldsymbol{v}\}\big]. \tag{2.15}$$

Consequently $\boldsymbol{\vartheta}_0$ is the unique minimizer of this contrast over Θ_s for the fixed latent representation. In $\mathcal{M}_0$, the analogous contrast formed with residuals $Y_i - \boldsymbol{Z}_i^\top \boldsymbol{\theta} - \boldsymbol{X}_i^\top \boldsymbol{\beta}$ identifies $(\boldsymbol{\theta}_0, \boldsymbol{\beta}_0)$ from the observed-data law, under the corresponding observed-design separation condition.

A proof is given in Appendix A. The lower bound uses ρ_A, which is quadratic for $|t| \le A$ and linear beyond that range. Section 4 develops the corresponding finite-sample bounds.

The primary observed-data target is $(\boldsymbol{\theta}_0, \boldsymbol{\beta}_0)$ in $\mathcal{M}_0$. Extension results for $\mathcal{M}_1$ apply to a specified factor-model sequence with explicit recovery conditions; identification across unrestricted latent decompositions requires further restrictions.

2.6 Centering and error criteria

Let $\boldsymbol{X}, \boldsymbol{F}, \boldsymbol{U}, \boldsymbol{Z}$ collect row vectors $\boldsymbol{X}_i^\top, \boldsymbol{f}_i^\top, \boldsymbol{u}_i^\top, \boldsymbol{Z}_i^\top$, respectively, and set

$$\begin{gathered} \boldsymbol{J} = \boldsymbol{I}_n - \mathbf{1}_n \mathbf{1}_n^\top / n, \\ \boldsymbol{X}_c = \boldsymbol{J}\boldsymbol{X}, \quad \boldsymbol{F}_c = \boldsymbol{J}\boldsymbol{F}, \quad \boldsymbol{U}_c = \boldsymbol{J}\boldsymbol{U}, \quad \boldsymbol{Z}_c = \boldsymbol{J}\boldsymbol{Z}. \end{gathered} \tag{2.16}$$

Then $\boldsymbol{X}_c = \boldsymbol{F}_c\boldsymbol{B}^\top + \boldsymbol{U}_c$, and

$$\begin{gathered}\boldsymbol{\eta}_c^0 := \boldsymbol{J}\boldsymbol{\eta}^0 = \boldsymbol{Z}_c\boldsymbol{\theta}_0 + \boldsymbol{U}_c\boldsymbol{\beta}_0 + \boldsymbol{F}_c\boldsymbol{\gamma}_0, \\ Y_i - (\eta_i^0 - \bar{\eta}^0) = R_i + \bar{\eta}^0.\end{gathered} \tag{2.17}$$

The common constant $\bar{\eta}^0$ cancels from each pairwise residual difference. Sample covariate centering therefore preserves the rank comparisons and slope target while accommodating arbitrary error location.

The primary error quantities are $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2$ and $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1$. The separate quantity $n^{-1/2}\|J(\widehat{\boldsymbol{\eta}} - \boldsymbol{\eta}^0)\|_2$ measures location-invariant prediction error on the training sample. Section 3 defines the feasible criterion and reference coordinates for these targets.

3 Estimation method

The estimator uses the observed data and the centering conventions of Section 2. Write $q = d + K$, $P = p + q$, $\boldsymbol{J} = \boldsymbol{I}_n - \mathbf{1}_n\mathbf{1}_n^\top/n$, $\boldsymbol{X}_c = \boldsymbol{J}\boldsymbol{X}$, and $\boldsymbol{Z}_c = \boldsymbol{J}\boldsymbol{Z}$. Its primary target is the sparse molecular slope in $\mathcal{M}_0$, with $\mathcal{M}_1$ interpreted relative to its specified latent representation.

3.1 Covariate-only factor estimation

Take $n \geq 2$ and a prespecified $K \leq \operatorname{rank}(\boldsymbol{X}_c) \leq n - 1$. For $K > 0$, let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_K$ be orthonormal eigenvectors of $\boldsymbol{X}_c\boldsymbol{X}_c^\top/n$ associated with its largest K eigenvalues $\widehat{\ell}_1 \geq \cdots \geq \widehat{\ell}_K > 0$, and define

$$\begin{aligned}\widehat{\boldsymbol{F}} &= \sqrt{n}\,(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_K), & \widehat{\boldsymbol{\Lambda}} &= \operatorname{diag}(\widehat{\ell}_1, \ldots, \widehat{\ell}_K), \\ \widehat{\boldsymbol{B}} &= \boldsymbol{X}_c^\top\widehat{\boldsymbol{F}}/n, & \widehat{\boldsymbol{U}} &= \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top.\end{aligned} \tag{3.1}$$

Signs and tied eigenspaces follow a deterministic covariate-measurable convention. A tie across the Kth eigenvalue can alter the selected subspace; the recovery analysis assumes spectral separation at K. The construction is outcome-free and gives

$$\begin{gathered}\widehat{\boldsymbol{F}}^\top\widehat{\boldsymbol{F}}/n = \boldsymbol{I}_K, \qquad \widehat{\boldsymbol{F}}^\top\widehat{\boldsymbol{U}} = 0, \qquad \boldsymbol{J}\widehat{\boldsymbol{F}} = \widehat{\boldsymbol{F}}, \quad \boldsymbol{J}\widehat{\boldsymbol{U}} = \widehat{\boldsymbol{U}}, \\ \widehat{\boldsymbol{B}}^\top\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{\Lambda}}, \qquad \boldsymbol{X}_c\widehat{\boldsymbol{B}} = \widehat{\boldsymbol{F}}\widehat{\boldsymbol{\Lambda}}\end{gathered} \tag{3.2}$$

directly. This principal-component residualization follows Fan et al. (2020); sample factor–residual orthogonality allows dependence among the residual coordinates.

Let $\widehat{\boldsymbol{f}}_i$ and $\widehat{\boldsymbol{u}}_i$ denote the column vectors corresponding to row i of $\widehat{\boldsymbol{F}}$ and $\widehat{\boldsymbol{U}}$. Set

$$\begin{aligned} \widehat{\boldsymbol{A}} &= (\boldsymbol{Z}_c, \widehat{\boldsymbol{F}}), \\ \widehat{\boldsymbol{W}} &= (\widehat{\boldsymbol{A}}, \widehat{\boldsymbol{U}}), \\ \boldsymbol{a} = (\boldsymbol{\theta}^\top, \boldsymbol{\gamma}^\top)^\top, &\quad \boldsymbol{\vartheta} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top. \end{aligned} \tag{3.3}$$

Thus $\widehat{\boldsymbol{W}}$ is $n \times P$, with row $\widehat{\boldsymbol{W}}_i^\top$. The convention $K = 0$ omits the factor block and sets $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c$, defining the unadjusted comparison method.

Residual coefficients retain the molecular coordinate scales fixed in Section 2. Under a rescaling $\widetilde{\boldsymbol{U}} = \widehat{\boldsymbol{U}}\boldsymbol{S}^{-1}$ with positive diagonal $\boldsymbol{S} = \mathrm{diag}(s_1, \ldots, s_p)$, the equivalent coordinates and penalty are $\boldsymbol{\zeta} = \boldsymbol{S}\boldsymbol{\beta}$ and $\lambda \sum_j |\zeta_j|/s_j$. All learned transformations are confined to the training sample.

3.2 Censored pairwise loss and scalar smoothing

For a finite parameter vector, define

$$\begin{aligned} e_i(\boldsymbol{\vartheta}) &= Y_i - \widehat{\boldsymbol{W}}_i^\top \boldsymbol{\vartheta}, \\ r_{ij}(\boldsymbol{\vartheta}) &= e_j(\boldsymbol{\vartheta}) - e_i(\boldsymbol{\vartheta}), \\ \boldsymbol{w}_{ij} &= \widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j, \\ N &= n(n-1). \end{aligned} \tag{3.4}$$

The unsmoothed Gehan loss is

$$L_{n,0}(\boldsymbol{\vartheta}) = \frac{1}{N} \sum_{i \neq j} \Delta_i \{ r_{ij}(\boldsymbol{\vartheta}) \}_+. \tag{3.5}$$

The criterion follows regularized Gehan estimation (Cai et al., 2009; Suder and Molstad, 2022). Event patients anchor the index i, and every other patient, including censored patients, contributes as j. We retain the ordered off-diagonal normalization $N = n(n-1)$ throughout.

Let $V \sim N(0,1)$ be an auxiliary integration variable, unrelated to the AFT error. For a common scalar bandwidth $h > 0$, define

$$\begin{aligned} \ell_h(r) &= E_V(r + hV)_+ = r\Phi(r/h) + h\phi(r/h), \\ L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{N}\sum_{i \neq j} \Delta_i \ell_h\{r_{ij}(\boldsymbol{\vartheta})\}, \end{aligned} \tag{3.6}$$

where ϕ and Φ are the standard normal density and distribution function. The Gaussian convolution is evaluated analytically while the AFT error distribution remains unspecified. Differentiation gives

$$\ell_h'(r) = \Phi(r/h), \qquad \ell_h''(r) = h^{-1}\phi(r/h) > 0. \tag{3.7}$$

Consequently ℓ_h is convex and 1-Lipschitz, with derivative $1/2$ at zero.

Proposition 3.1 (Uniform loss approximation)**.** *For every $r \in \mathbb{R}$ and $h > 0$,*

$$\begin{aligned} 0 \leq \ell_h(r) - r_+ &\leq \frac{h}{\sqrt{2\pi}}, \\ \int_{\mathbb{R}} |\Phi(r/h) - \mathbf{1}\{r \geq 0\}| \, \mathrm{d}r &= h\sqrt{\frac{2}{\pi}}. \end{aligned} \tag{3.8}$$

For any fixed observed sample, writing $\bar{\Delta} = n^{-1}\sum_i \Delta_i$,

$$0 \leq L_{n,h}(\boldsymbol{\vartheta}) - L_{n,0}(\boldsymbol{\vartheta}) \leq \frac{\bar{\Delta} h}{\sqrt{2\pi}} \quad \textit{uniformly over } \boldsymbol{\vartheta} \in \mathbb{R}^P. \tag{3.9}$$

Appendix B proves the deterministic identities in this section. Proposition 3.1 controls the loss approximation; Section 4 combines bandwidth-dependent score bias, identification, and curvature to obtain coefficient bounds.

Scalar smoothing of censored rank scores has precedents in Heller (2007). It uses a common bandwidth, whereas the parameter-perturbation construction of Johnson and Strawderman (2009) induces pair-specific bandwidths. For a perturbation $n^{-1/2}\boldsymbol{\Gamma}\boldsymbol{V}_P$, with $\boldsymbol{V}_P \sim N(0, \boldsymbol{I}_P)$, these are

$$h_{ij} = n^{-1/2}\{\boldsymbol{w}_{ij}^\top \boldsymbol{\Gamma}\boldsymbol{\Gamma}^\top \boldsymbol{w}_{ij}\}^{1/2},$$

which generally varies by pair. The common scalar h used here leads to the smoothing-bias analysis in Section 4.

3.3 Partial regularization and finite minimizers

For fixed $h > 0$ and $\lambda > 0$, let

$$\begin{aligned} \mathcal{S}_{n,h,\lambda} &= \arg\min_{\vartheta \in \mathbb{R}^P} Q_{n,h,\lambda}(\boldsymbol{\vartheta}), \\ Q_{n,h,\lambda}(\boldsymbol{\vartheta}) &= L_{n,h}(\boldsymbol{\vartheta}) + \lambda \|\boldsymbol{\beta}\|_1. \end{aligned} \tag{3.10}$$

When this set is nonempty, $\widehat{\boldsymbol{\vartheta}}$ denotes a member. The ℓ_1 penalty produces sparsity in the p molecular coefficients, and clinical and factor effects are estimated jointly with them. Section 4 establishes coefficient-error guarantees.

The identity $\widehat{\boldsymbol{U}} = \boldsymbol{X}_c - \widehat{\boldsymbol{F}}\widehat{\boldsymbol{B}}^\top$ implies

$$\begin{aligned} \widehat{\boldsymbol{W}}\boldsymbol{\vartheta} &= \boldsymbol{Z}_c\boldsymbol{\theta} + \boldsymbol{X}_c\boldsymbol{\beta} + \widehat{\boldsymbol{F}}\boldsymbol{\nu}, \\ \boldsymbol{\nu} &= \boldsymbol{\gamma} - \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}. \end{aligned} \tag{3.11}$$

For fixed training data, (3.11) is a bijective reparameterization preserving $\|\boldsymbol{\beta}\|_1$. The observed-$\boldsymbol{X}$ formulation includes a freely estimated principal-component term. Setting $\boldsymbol{\gamma} = \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}$ recovers the ordinary penalized smoothed AFT fit; allowing $\boldsymbol{\gamma}$ to vary gives the proposed relaxation. Both blocks are optimized jointly under the nonlinear pairwise loss.

Finite attainment with the unpenalized block is governed by the nuisance-design geometry. A sufficient sample condition, when $q > 0$, is

$$\chi_n := \inf_{\|\boldsymbol{v}\|_2 = 1} \frac{1}{N} \sum_{i \neq j} \Delta_i \{(\widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_j)^\top \boldsymbol{v}\}_+ > 0. \tag{3.12}$$

Condition (3.12) makes $\mathcal{S}_{n,h,\lambda}$ nonempty and compact; Appendix B.3 proves this result and illustrates nonattainment. For $q = 0$, the positive ℓ_1 penalty bounds the sublevel sets. Section 4 also establishes finite attainment on the statistical event used for estimation bounds.

A deterministic tie-breaking rule may select among multiple minimizers; numerical eligibility is recorded separately.

3.4 Reference coordinates and the approximation term

To relate (3.10) to the target, write the centered latent matrices as $\boldsymbol{F}_c = \boldsymbol{J}\boldsymbol{F}$ and $\boldsymbol{U}_c = \boldsymbol{J}\boldsymbol{U}$. For any invertible alignment matrix $\boldsymbol{H}$, set

$$\begin{aligned} \boldsymbol{E}_F &= \widehat{\boldsymbol{F}} - \boldsymbol{F}_c\boldsymbol{H}^\top, \\ \boldsymbol{\gamma}^\dagger &= \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}_0 + \boldsymbol{H}^{-T}\boldsymbol{\xi}_0, \\ \boldsymbol{\vartheta}^\dagger &= (\boldsymbol{\theta}_0^\top, (\boldsymbol{\gamma}^\dagger)^\top, \boldsymbol{\beta}_0^\top)^\top. \end{aligned} \tag{3.13}$$

These reference coordinates link the feasible criterion to the population target. In $\mathcal{M}_0$, $\boldsymbol{\gamma}^\dagger = \widehat{\boldsymbol{B}}^\top\boldsymbol{\beta}_0$ and the alignment matrix drops out. Recall

$$\boldsymbol{\eta}^0 = \boldsymbol{Z}\boldsymbol{\theta}_0 + \boldsymbol{X}\boldsymbol{\beta}_0 + \boldsymbol{F}\boldsymbol{\xi}_0, \qquad \boldsymbol{\eta}_c^0 = \boldsymbol{J}\boldsymbol{\eta}^0, \qquad R_i = Y_i - \eta_i^0.$$

Proposition 3.2 (Exact reference representation)**.** *With* $\mathbf{d} = \boldsymbol{E}_F\boldsymbol{H}^{-T}\boldsymbol{\xi}_0$ *and* $b_n = \|\mathbf{d}\|_\infty$*,*

$$\begin{aligned} \widehat{\boldsymbol{W}}\boldsymbol{\vartheta}^\dagger &= \boldsymbol{\eta}_c^0 + \mathbf{d}, & \boldsymbol{J}\mathbf{d} &= \mathbf{d}, \\ e_i(\boldsymbol{\vartheta}^\dagger) &= R_i + \bar{\eta}^0 - d_i, & r_{ij}(\boldsymbol{\vartheta}^\dagger) &= R_j - R_i + d_i - d_j. \end{aligned} \tag{3.14}$$

Moreover,

$$\begin{aligned} b_n &\le \left(\max_i \|(\boldsymbol{E}_F)_{i\cdot}\|_2\right) \|\boldsymbol{H}^{-T}\boldsymbol{\xi}_0\|_2, \\ \widehat{\boldsymbol{W}}\hat{\boldsymbol{\vartheta}} - \boldsymbol{\eta}_c^0 &= \widehat{\boldsymbol{W}}(\hat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger) + \mathbf{d}. \end{aligned} \tag{3.15}$$

In $\mathcal{M}_0$*,* $\mathbf{d} = 0$ *exactly for every admissible PCA output.*

The common location in (3.14) cancels within each pair. Thus b_n measures approximation of the additional factor contribution. For $\mathcal{M}_0$, the ordinary predictor is represented exactly and factor estimation enters through the estimated-design geometry. Nuisance error is measured from the sample-dependent reference $\boldsymbol{\gamma}^\dagger$, allowing $\boldsymbol{B}^\top\boldsymbol{\beta}_0$ to grow with the sparse signal.

Within a fixed estimated subspace, replacing $\widehat{\boldsymbol{F}}$ and $\widehat{\boldsymbol{B}}$ by $\widehat{\boldsymbol{F}}\boldsymbol{Q}^\top$ and $\widehat{\boldsymbol{B}}\boldsymbol{Q}^\top$, and $\boldsymbol{\gamma}$ by $\boldsymbol{Q}\boldsymbol{\gamma}$, preserves $\widehat{\boldsymbol{U}}$, the predictor, and the objective. The fitted-predictor and molecular-minimizer sets are invariant under this coordinate rotation. Subspace selection at a spectral tie is governed by the convention in Section 3.1.

3.5 Training-frozen scores for held-out patients

For a new patient, the training means and loadings define a frozen prediction map. Let $\boldsymbol{x}_c = \boldsymbol{x} - \bar{\boldsymbol{X}}$ and $\boldsymbol{z}_c = \boldsymbol{z} - \bar{\boldsymbol{Z}}$. For $K > 0$, set

$$\begin{aligned} \widehat{\boldsymbol{f}}(\boldsymbol{x}) &= \widehat{\boldsymbol{\Lambda}}^{-1}\widehat{\boldsymbol{B}}^{\top}\boldsymbol{x}_c, \\ \widehat{\boldsymbol{u}}(\boldsymbol{x}) &= \boldsymbol{x}_c - \widehat{\boldsymbol{B}}\widehat{\boldsymbol{f}}(\boldsymbol{x}), \\ \widehat{\eta}(\boldsymbol{x},\boldsymbol{z}) &= \boldsymbol{z}_c^{\top}\widehat{\boldsymbol{\theta}} + \widehat{\boldsymbol{f}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\gamma}} + \widehat{\boldsymbol{u}}(\boldsymbol{x})^{\top}\widehat{\boldsymbol{\beta}}. \end{aligned} \tag{3.16}$$

For $K = 0$, omit the factor terms. By (3.2), this least-squares projection onto the estimated loading space reproduces the fitted training factors and residuals exactly.

Equivalently,

$$\begin{aligned} \widehat{\eta}(\boldsymbol{x},\boldsymbol{z}) &= \boldsymbol{z}_c^{\top}\widehat{\boldsymbol{\theta}} + \boldsymbol{x}_c^{\top}\widehat{\boldsymbol{b}}_{\mathrm{pred}}, \\ \widehat{\boldsymbol{b}}_{\mathrm{pred}} &= \widehat{\boldsymbol{\beta}} + \widehat{\boldsymbol{B}}\widehat{\boldsymbol{\Lambda}}^{-1}(\widehat{\boldsymbol{\gamma}} - \widehat{\boldsymbol{B}}^{\top}\widehat{\boldsymbol{\beta}}). \end{aligned} \tag{3.17}$$

The sparse coefficient $\widehat{\boldsymbol{\beta}}$ summarizes residual effects, whereas $\widehat{\boldsymbol{b}}_{\mathrm{pred}}$ combines residual and factor-mediated contributions and can be dense. Higher $\widehat{\eta}$ denotes longer conditional event times; $-\widehat{\eta}$ supplies the opposite risk ordering. The score represents relative log-time location. Calibrated event-time quantiles additionally require the error distribution and its location.

3.6 Bandwidth, penalty, and training selection

The estimator is indexed by (K, h, λ). The coefficient analysis takes the correct K as fixed, with $h = h_n > 0$ and $\lambda = \lambda_n > 0$ deterministic or fixed after conditioning on the design. Section 4 relates the smoothing bias, b_n, and stochastic fluctuations to the penalty and gives sufficient bandwidth and penalty conditions for the stated error rates.

For empirical tuning, let $\mathcal{V}_1, \ldots, \mathcal{V}_L$ be held-out folds of sizes $m_\ell \geq 2$. For every candidate (K, h, λ), all learned preprocessing, PCA, and coefficient fitting use the complement of $\mathcal{V}_\ell$. Apply (3.16)

to obtain $\widehat{\eta}_i^{(-\ell)}$ and put $e_i^{(-\ell)} = Y_i - \widehat{\eta}_i^{(-\ell)}$. The cross-validation (CV) score is

$$\mathrm{CV}(K,h,\lambda) = \frac{\displaystyle\sum_{\ell=1}^{L} \sum_{\substack{i,j\in\mathcal{V}_\ell \\ i\neq j}} \Delta_i \left[\{e_j^{(-\ell)} - e_i^{(-\ell)}\}_+ - \{Y_j - Y_i\}_+\right]}{\displaystyle\sum_{\ell=1}^{L} m_\ell(m_\ell - 1)}. \tag{3.18}$$

The candidate-independent baseline subtraction preserves the selection while expressing validation as a finite loss contrast. The same unsmoothed score is used for every candidate bandwidth.

Within-fold differences remove arbitrary fold-specific locations. All candidate bandwidths share the same unsmoothed score, computed over within-fold pairs. Alternative pooling rules are studied by Suder and Molstad (2022). Section 6.4 specifies numerical eligibility and tie handling for the empirical study. The coefficient theorems use fixed or covariate-measurable tuning; the outcome-selected procedure is evaluated separately.

3.7 Derivative and optimality interface

For $h > 0$, differentiation of the finite sum gives

$$\begin{aligned} \nabla L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{N} \sum_{i\neq j} \Delta_i \Phi\{r_{ij}(\boldsymbol{\vartheta})/h\} \boldsymbol{w}_{ij}, \\ \nabla^2 L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{Nh} \sum_{i\neq j} \Delta_i \phi\{r_{ij}(\boldsymbol{\vartheta})/h\} \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^\top. \end{aligned} \tag{3.19}$$

The positive-semidefinite Hessian establishes convexity. A finite vector is a global minimizer of (3.10) if and only if

$$\begin{aligned} \nabla_a L_{n,h}(\widehat{\boldsymbol{\vartheta}}) &= 0, \\ 0 &\in \nabla_\beta L_{n,h}(\widehat{\boldsymbol{\vartheta}}) + \lambda \partial \|\widehat{\boldsymbol{\beta}}\|_1. \end{aligned} \tag{3.20}$$

A nonzero molecular coordinate has gradient $-\lambda \operatorname{sign}(\widehat{\beta}_j)$, and a zero coordinate has gradient in $[-\lambda, \lambda]$. Numerical stationarity is assessed on these coordinates and the complete unpenalized block.

The gradient admits an efficient matrix form. Define $V_{ii} = 0$, $V_{ij} = \Delta_i \Phi(r_{ij}/h)$ for $i \neq j$, and $c_i = \sum_j V_{ij} - \sum_j V_{ji}$. Then

$$\nabla L_{n,h}(\boldsymbol{\vartheta}) = \widehat{\boldsymbol{W}}^\top \boldsymbol{c}/N. \tag{3.21}$$

4 Theoretical properties

The analysis combines conditional score control, local event information, and estimated-design geometry to establish finite-sample coefficient bounds. For the ordinary model $\mathcal{M}_0$, the reference predictor is exact. Primitive factor conditions then supply the design event, and the extension to $\mathcal{M}_1$ adds a separate approximation term. Pairing and contraction (Clémençon et al., 2008) and localized regularization (Negahban et al., 2012) provide the analytic tools.

4.1 Conditional setup and design geometry

Write $q = d + K$, $P = p + q$, $k = s + q \geq 1$, $N = n(n-1)$, and condition on $D_n = \sigma(\boldsymbol{D}_1, \ldots, \boldsymbol{D}_n)$, where $\boldsymbol{D}_i = (\boldsymbol{Z}_i^\top, \boldsymbol{f}_i^\top, \boldsymbol{u}_i^\top)^\top$. The factor estimates, centering, and tuning in this analysis are covariate-measurable. Outcomes are conditionally independent with patient-specific conditional laws. The criterion remains

$$\begin{aligned} Q(\boldsymbol{\vartheta}) &= L_{n,h}(\boldsymbol{\vartheta}) + \lambda \|\boldsymbol{\beta}\|_1, \\ L_{n,h}(\boldsymbol{\vartheta}) &= \frac{1}{N} \sum_{i \neq j} \Delta_i \ell_h\{Y_j - Y_i + \boldsymbol{w}_{ij}^\top \boldsymbol{\vartheta}\}, \end{aligned} \tag{4.1}$$

where $\ell_h(t) = t\Phi(t/h) + h\phi(t/h)$, $\widehat{\boldsymbol{W}} = (\boldsymbol{Z}_c, \widehat{\boldsymbol{F}}, \widehat{\boldsymbol{U}})$ is centered, and $\boldsymbol{w}_{ij} = \widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j$. In $\mathcal{M}_0$, the reference point is

$$\begin{aligned} \boldsymbol{\vartheta}^\dagger &= (\boldsymbol{\theta}_0^\top, (\widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0)^\top, \boldsymbol{\beta}_0^\top)^\top, \\ \widehat{\boldsymbol{W}} \boldsymbol{\vartheta}^\dagger &= \boldsymbol{\eta}_c^0. \end{aligned} \tag{4.2}$$

Conditioning fixes the sample-dependent reference. Its representation accommodates unrestricted $\|\boldsymbol{B}^\top \boldsymbol{\beta}_0\|_2$.

Two sample scales are kept distinct:

$$V_n = \max_{j \le P} \frac{\|\widehat{\boldsymbol{W}}_{\cdot j}\|_2}{\sqrt{n}}, \qquad M_n = \max_{i,j} |\widehat{W}_{ij}|, \qquad \widehat{\boldsymbol{\Gamma}} = \frac{1}{N} \sum_{i \neq j} \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^\top = \frac{2}{n-1} \widehat{\boldsymbol{W}}^\top \widehat{\boldsymbol{W}}. \tag{4.3}$$

The column scale V_n controls stochastic fluctuations and score bias. The larger entry envelope M_n is used only to keep predictors inside a common local information interval. For $\boldsymbol{v} = (\boldsymbol{v}_a^\top, \boldsymbol{v}_\beta^\top)^\top$, define

$$\mathcal{C}(S) = \{\boldsymbol{v} : \|(\boldsymbol{v}_\beta)_{S^c}\|_1 \le 3(\|(\boldsymbol{v}_\beta)_S\|_1 + \|\boldsymbol{v}_a\|_1)\}. \tag{4.4}$$

On this cone, $\|\boldsymbol{v}\|_1 \le 4\sqrt{k}\|\boldsymbol{v}\|_2$. Thus the unpenalized block contributes to the effective dimension.

Assumption 4.1 (Conditional restricted design)**.** The covariate realization satisfies, for some $\kappa_X > 0$,

$$\boldsymbol{v}^\top \widehat{\boldsymbol{\Gamma}} \boldsymbol{v} \ge \kappa_X \|\boldsymbol{v}\|_2^2 \quad \text{for every } \boldsymbol{v} \in \mathcal{C}(S). \tag{4.5}$$

Assumption 4.1 controls the estimated design on the error cone $\mathcal{C}(S)$, the geometry relevant to penalized estimation.

Assumptions 2.1–2.2 specify the common error law, conditional censoring, bounded error density, and a design-uniform local information interval. They permit censoring atoms and log-time errors without moments. Put

$$c_{\mathrm{loc}} = A m_\varepsilon^2 g_0^2, \qquad c_\phi = 2\Phi(1) - 1, \qquad \kappa = c_{\mathrm{loc}} c_\phi \kappa_X. \tag{4.6}$$

4.2 Score bias, pairwise fluctuations, and curvature

Let $R_i = Y_i - \eta_i^0$. To cover both models in a single set of analytic lemmas, allow a covariate-measurable centered vector $\mathbf{d}$ with $\|\mathbf{d}\|_\infty \le b$, such that

$$\widehat{\boldsymbol{W}} \boldsymbol{\vartheta}^\dagger = \boldsymbol{\eta}_c^0 + \mathbf{d}, \qquad r_{ij}(\boldsymbol{\vartheta}^\dagger) = R_j - R_i + d_i - d_j.$$

For $\mathcal{M}_0$, take $b = 0$. Define the conditional risk *contrast* and mean score by

$$\begin{aligned} \mathcal{R}_h(\boldsymbol{v}) &= E\{L_{n,h}(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^\dagger) \mid D_n\}, \\ \boldsymbol{g}_h &= E\{\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger) \mid D_n\}. \end{aligned} \tag{4.7}$$

The loss increment is bounded in absolute value by $2V_n\|\boldsymbol{v}\|_1$, making (4.7) integrable without an error-moment assumption.

Lemma 4.1 (Score and contrast control)**.** *Suppose the common-error and conditional-censoring assumptions hold. Fix* $0 < \alpha < 1/4$ *and* $r > 0$ *after conditioning, and set*

$$t_\alpha = \sqrt{\frac{\log(2P/\alpha)}{n}}, \qquad \mu(b,h) = 2V_n L_\varepsilon\{2b + h\sqrt{2/\pi}\}. \tag{4.8}$$

Then $\|\boldsymbol{g}_h\|_\infty \le \mu(b,h)$*. Each of the following inequalities holds with conditional probability at least* $1-\alpha$*:*

$$\|\nabla L_{n,h}(\boldsymbol{\vartheta}^\dagger) - \boldsymbol{g}_h\|_\infty \le \sqrt{8}\, V_n t_\alpha, \tag{4.9}$$

$$\sup_{\|\boldsymbol{v}\|_1 \le r} |L_{n,h}(\boldsymbol{\vartheta}^\dagger + \boldsymbol{v}) - L_{n,h}(\boldsymbol{\vartheta}^\dagger) - \mathcal{R}_h(\boldsymbol{v})| \le 96 V_n r t_\alpha. \tag{4.10}$$

The two events hold jointly with probability at least $1-2\alpha$*.*

The stochastic scale is $V_n\sqrt{\log(2P/\alpha)/n}$: patients are the independent sampling units, and the bound is uniform in h. Averaging over all pairings retains patient-specific censoring laws. Rademacher moment contraction (Clémençon et al., 2008; Ledoux and Talagrand, 1991) yields column-scale control in Appendix C.3.

Lemma 4.2 (Conditional local curvature)**.** *Under the outcome conditions and Assumption 4.1, if* $h \le A/2$ *and*

$$2b + 2M_n\|\boldsymbol{v}\|_1 \le A/2, \tag{4.11}$$

then, for $\boldsymbol{v} \in \mathcal{C}(S)$*,*

$$\mathcal{R}_h(\boldsymbol{v}) - \boldsymbol{g}_h^\top \boldsymbol{v} \ge \frac{c_{\mathrm{loc}} c_\phi}{2} \boldsymbol{v}^\top \widehat{\boldsymbol{\Gamma}} \boldsymbol{v} \ge \frac{\kappa}{2}\|\boldsymbol{v}\|_2^2. \tag{4.12}$$

Lemma 4.2 supplies curvature of the conditional expected contrast on $\mathcal{C}(S)$, while Lemma 4.1 controls its empirical fluctuation. The event-weighted difference density permits censoring-induced atoms.

4.3 Ordinary-model estimation and existence

Theorem 4.1 (Conditional guarantee in $\mathcal{M}_0$)**.** *Suppose $n \geq 2$, the ordinary model and outcome conditions hold, and the conditioned design satisfies Assumption 4.1. For $0 < \alpha < 1/4$, choose $h > 0$, $\lambda > 0$, and*

$$\lambda \geq 4\mu(0, h) + 384 V_n t_\alpha, \qquad R = \frac{8\sqrt{k}\lambda}{\kappa}. \tag{4.13}$$

If

$$h \leq A/2, \qquad 8M_n\sqrt{k}R \leq A/2, \tag{4.14}$$

then, with conditional probability at least $1 - 2\alpha$, the set of global minimizers of Q is nonempty and compact. Every such minimizer satisfies

$$\begin{aligned} \|\widehat{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger\|_2 &\leq \frac{8\sqrt{k}\lambda}{\kappa}, \\ \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 &\leq \frac{32k\lambda}{\kappa}. \end{aligned} \tag{4.15}$$

In particular the first bound also controls $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2$.

Theorem 4.1 establishes finite attainment and coefficient bounds on the same statistical event. Appendix C.5 localizes the sublevel set through $\boldsymbol{\vartheta}^\dagger$ inside a compact ball.

The localized basic inequality follows the decomposable-regularization framework (Negahban et al., 2012) with a seminorm penalty on the full parameter. Including the nuisance directions yields the sharper cone relation

$$\|(\boldsymbol{v}_\beta)_{S^c}\|_1 \leq 3\|(\boldsymbol{v}_\beta)_S\|_1 + \|\boldsymbol{v}_a\|_1. \tag{4.16}$$

The nuisance error is $\widehat{\boldsymbol{\gamma}} - \widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0$, measured in the estimated-factor coordinates. The sparse molecular coefficient and the deployment slope retain their distinct definitions from Section 3.

4.4 From primitive factor conditions to estimated-design geometry

The primitive conditions concern the joint covariate distribution. They supply the estimated-design geometry separately from the assumptions on ε_i.

Assumption 4.2 (Independent jointly sub-Gaussian factor design)**.** For fixed $K \geq 1$ and $d \geq 0$, the centered $\boldsymbol{D}_i$ are independent and identically distributed and satisfy

$$\begin{aligned} c_D \boldsymbol{I}_P \preceq \boldsymbol{\Sigma}_D := E(\boldsymbol{D}_i \boldsymbol{D}_i^\top) \preceq C_D \boldsymbol{I}_P, \\ \sup_{\|\boldsymbol{v}\|_2 = 1} \|\boldsymbol{v}^\top \boldsymbol{D}_i\|_{\psi_2} \leq L_D. \end{aligned} \tag{4.17}$$

Here $\|U\|_{\psi_2} = \inf\{t > 0 : E \exp(U^2/t^2) \leq 2\}$. Also $E(\boldsymbol{f}_i \boldsymbol{f}_i^\top) = \boldsymbol{I}_K$, $E(\boldsymbol{f}_i \boldsymbol{u}_i^\top) = 0$, and the deterministic loading matrix obeys

$$c_B \boldsymbol{I}_K \preceq \boldsymbol{B}^\top \boldsymbol{B}/p \preceq C_B \boldsymbol{I}_K, \qquad \max_{j \leq p} \|\boldsymbol{B}_{j\cdot}\|_2 \leq L_B. \tag{4.18}$$

All displayed constants are uniform in n, p.

The covariance lower bound and factor pervasiveness jointly specify a sufficient design regime. The normalization and alignment follow the PCA framework of Bai and Ng (2013) and Fan et al. (2020); Assumption 4.2 supports the norm bounds below.

For $0 < \delta < 1/4$, put

$$\begin{aligned} &L_\delta = \log(64nP/\delta), \quad t_\delta = \sqrt{L_\delta/n}, \quad \tau_\delta = t_\delta + L_\delta/n, \\ &a_\delta = \tau_\delta + p^{-1/2}, \quad r_{F,\delta} = n^{-1/2} + \sqrt{L_\delta/p}. \end{aligned} \tag{4.19}$$

The quantity χ_n in Section 3 continues to denote the separate sample-existence criterion.

Proposition 4.1 (PCA and design event)**.** *Under Assumption 4.2, there are fixed positive constants c, C such that, if $n \wedge p \geq C L_\delta$, an event $\mathcal{E}_D$ has probability at least $1 - \delta$ and the following properties. The selected spectrum is separated at K, $cp\boldsymbol{I}_K \preceq \widehat{\boldsymbol{\Lambda}} \preceq Cp\boldsymbol{I}_K$, and*

$$\begin{aligned} \boldsymbol{H} = \widehat{\boldsymbol{\Lambda}}^{-1} \widehat{\boldsymbol{B}}^\top \boldsymbol{B}, \\ \boldsymbol{E}_F = \widehat{\boldsymbol{F}} - \boldsymbol{F}_c \boldsymbol{H}^\top = \boldsymbol{U}_c \widehat{\boldsymbol{B}} \widehat{\boldsymbol{\Lambda}}^{-1}, \\ \|\boldsymbol{H}\|_{\mathrm{op}} + \|\boldsymbol{H}^{-1}\|_{\mathrm{op}} \leq C. \end{aligned} \tag{4.20}$$

On this event,

$$\begin{aligned} \|\boldsymbol{E}_F\|_{\mathrm{op}}/\sqrt{n} \leq C(n^{-1/2} + p^{-1/2}), \\ \max_i \|(\boldsymbol{E}_F)_{i\cdot}\|_2 \leq C r_{F,\delta}. \end{aligned} \tag{4.21}$$

$$\begin{aligned} \max_j \|\widehat{\boldsymbol{U}}_{\cdot j} - \boldsymbol{U}_{c,\cdot j}\|_2/\sqrt{n} \le C a_\delta, \\ V_n \le C, \qquad M_n \le C\sqrt{L_\delta}. \end{aligned} \tag{4.22}$$

If, additionally,

$$k\tau_\delta \le c, \qquad \sqrt{k}a_\delta \le c, \tag{4.23}$$

then Assumption 4.1 holds with a fixed lower bound $\kappa_X \ge c_X > 0$. If Assumption 4.2 is supplemented by $\max_j |D_{ij}| \le L_0$ almost surely for a fixed L_0, then $M_n \le C$ on the same type of event.

The operator, rowwise, and columnwise bounds support factor alignment, predictor approximation, and design transfer, respectively. Appendix D derives them through covariance concentration and sub-Gaussian net bounds (Vershynin, 2012), retaining the same-sample dependence of $\widehat{\boldsymbol{F}}$ and $\boldsymbol{U}_c$.

For clarity, the deterministic transfer step is stated separately. Let $\boldsymbol{W} = (\boldsymbol{Z}_c, \boldsymbol{F}_c\boldsymbol{H}^\top, \boldsymbol{U}_c)$ and define $\|\boldsymbol{x}\|_{\text{pair}}^2 = N^{-1}\sum_{i\neq j}(x_i - x_j)^2$.

Lemma 4.3 (Column-error transfer)**.** *If $\max_j \|\widehat{\boldsymbol{W}}_{\cdot j} - \boldsymbol{W}_{\cdot j}\|_2/\sqrt{n} \le a$ and $\|\boldsymbol{W}\boldsymbol{v}\|_{\text{pair}} \ge \sqrt{\kappa_0}\|\boldsymbol{v}\|_2$ for $\boldsymbol{v} \in \mathcal{C}(S)$, then*

$$\begin{aligned} \|\widehat{\boldsymbol{W}}\boldsymbol{v}\|_{\text{pair}} \ge (\sqrt{\kappa_0} - 8a\sqrt{k})\|\boldsymbol{v}\|_2, \\ \boldsymbol{v} \in \mathcal{C}(S). \end{aligned} \tag{4.24}$$

Thus $8a\sqrt{k} \le \sqrt{\kappa_0}/2$ yields a positive estimated-design lower bound. Column mean-square control suffices for this restricted-design transfer.

4.5 Unconditional ordinary-model rates

Theorem 4.2 (A primitive-condition guarantee for $\mathcal{M}_0$)**.** *Suppose the ordinary model, the outcome conditions, and Assumption 4.2 hold with uniform constants. There are constants $C_*, c_* > 0$, depending only on these constants and fixed K, d, for which the following statement holds. Take $0 < \delta < 1/4$, $n \wedge p \ge C_* L_\delta$, and impose (4.23) with sufficiently small constants. Choose*

$$0 < h \le A/2, \qquad \lambda = C_*\{t_\delta + h\}, \qquad k\sqrt{L_\delta}\,\lambda \le c_*. \tag{4.25}$$

Then, with probability at least $1-3\delta$*, the minimizer set is nonempty and compact, and every minimizer satisfies*

$$\begin{aligned} \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2 &\le C\sqrt{k}(t_\delta + h), \\ \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 &\le Ck(t_\delta + h). \end{aligned} \tag{4.26}$$

The corresponding reference-coordinate full-vector bound also holds. Under the additional bounded-coordinate condition of Proposition 4.1, the final condition in (4.25) *can instead be* $k\lambda \le c_*$.

With C_* sufficiently large and c_* sufficiently small, the tuning conditions combine smoothing bias and sampling fluctuations. The probability accounts for the design event and both conditional outcome events. Taking $h \le Ct_\delta$ gives the ℓ_2 rate $\sqrt{kL_\delta/n}$ in both covariate classes. For $p \ge n$, sufficient growth conditions are

$$\begin{aligned} &k^2 L_\delta^2/n \longrightarrow 0, \text{ jointly sub-Gaussian design,} \\ &k^2 L_\delta/n \longrightarrow 0, \text{ jointly sub-Gaussian and bounded coordinates.} \end{aligned} \tag{4.27}$$

For a probability tending to one, take $\delta = \delta_n \downarrow 0$ in these sufficient growth conditions. Using the column scale V_n in the stochastic bound removes an additional logarithmic loss from the error rate; the entry envelope M_n remains in the locality requirement.

A concrete construction demonstrates compatibility with Cauchy log-time errors. Take fixed $K = 1$, independent Rademacher coordinates in $(\boldsymbol{Z}_i, \boldsymbol{f}_i, \boldsymbol{u}_i)$, and $\boldsymbol{B}_j = b_0 \ne 0$. Let the error be standard Cauchy and independent of the design, and set $\log C_i = \eta_i^0 + 2$. On $[-1, 1]$, the error density is positive and $G = 1$, while censoring has positive probability. With $p = n^2$ and $s = \lfloor n^{1/4} \rfloor$, the bounded-design condition holds for polynomially decreasing δ_n.

4.6 Additional factor effects

In $\mathcal{M}_1$, fix the latent representation described in Section 2. For a covariate-measurable invertible $\boldsymbol{H}$, let

$$\begin{aligned} \boldsymbol{\gamma}^\dagger &= \widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0 + \boldsymbol{H}^{-T} \boldsymbol{\xi}_0, \\ \mathbf{d} &= \boldsymbol{E}_F \boldsymbol{H}^{-T} \boldsymbol{\xi}_0, \\ b_n &= \|\mathbf{d}\|_\infty. \end{aligned} \tag{4.28}$$

These reference quantities isolate approximation of the additional factor contribution.

Corollary 4.1 (Conditional and primitive-design extensions)**.** *Theorem 4.1 holds for this reference point after replacing* $\mu(0, h)$ *by* $\mu(b_n, h)$ *and replacing* (4.14) *by*

$$h \le A/2, \qquad 2b_n + 8M_n\sqrt{k}R \le A/2. \tag{4.29}$$

Under the primitive design assumptions and $\|\boldsymbol{\xi}_0\|_2 \le \Xi$*, Proposition 4.1 yields* $b_n \le C\Xi r_{F,\delta}$*. In the unconditional statement take*

$$\lambda = C_*\{t_\delta + h + \Xi r_{F,\delta}\}, \qquad \Xi r_{F,\delta} + k\sqrt{L_\delta}\lambda \le c_*. \tag{4.30}$$

The ℓ_2 *and* ℓ_1 *bounds are respectively*

$$C\sqrt{k}\{t_\delta + h + \Xi r_{F,\delta}\}, \qquad Ck\{t_\delta + h + \Xi r_{F,\delta}\}, \tag{4.31}$$

with probability at least $1 - 3\delta$*. With bounded coordinates replace* $k\sqrt{L_\delta}\lambda$ *by* $k\lambda$*. Other primitive-design and bandwidth conditions remain in force.*

Setting $\boldsymbol{\xi}_0 = 0$ deletes the approximation term exactly. For $p \ge n$ and bounded Ξ, $r_{F,\delta} \lesssim t_\delta$. These extension bounds retain the specified latent representation.

The coefficient guarantees apply to fixed, correct K and covariate-measurable tuning under the stated error, censoring, and design conditions. Section 5 connects them to finite numerical accuracy; Section 6 studies the separate outcome-based selection procedure.

5 Computation

We solve the partially penalized smoothed-rank objective in the original coefficient coordinates. For a fixed training sample and

(K, h, λ) with $h, \lambda > 0$, write $\widehat{\boldsymbol{W}} = (\widehat{\boldsymbol{A}}, \widehat{\boldsymbol{U}})$, $\widehat{\boldsymbol{A}} = (\boldsymbol{Z}_c, \widehat{\boldsymbol{F}})$, $q = d+K$, $P = q + p$, and $\boldsymbol{\vartheta} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top$. The objective is

$$\begin{aligned} Q(\boldsymbol{\vartheta}) &= f_h(\boldsymbol{\vartheta}) + g_\lambda(\boldsymbol{\vartheta}), \\ g_\lambda(\boldsymbol{\vartheta}) &= \lambda \|\boldsymbol{\beta}\|_1, \\ f_h(\boldsymbol{\vartheta}) &= \frac{1}{N} \sum_{i \neq j} \Delta_i \ell_h(r_{ij}), \\ r_{ij} &= Y_j - Y_i + (\widehat{\boldsymbol{W}}_i - \widehat{\boldsymbol{W}}_j)^\top \boldsymbol{\vartheta}, \\ N &= n(n-1). \end{aligned} \tag{5.1}$$

where Y_i is already on the log-time scale and $\ell_h(r) = r\Phi(r/h) + h\phi(r/h)$. All arrays are finite, $n \geq 2$, and $\Delta_i \in \{0, 1\}$. PCA and preprocessing remain fixed during a solve.

The reference algorithm is monotone proximal gradient with backtracking. The empirical implementation combines acceleration and qualified second-order refinement. These algorithms specialize composite-convex optimization (Beck and Teboulle, 2009; Parikh and Boyd, 2014) to the pairwise loss and unpenalized block in (5.1). Scalable alternatives for the unsmoothed AFT criterion include the prox-linear alternating direction method of multipliers (ADMM) algorithm of Suder and Molstad (2022).

5.1 Blocked pairwise evaluation

Compute $\boldsymbol{e} = \boldsymbol{Y} - \widehat{\boldsymbol{W}}\boldsymbol{\vartheta}$. For $i \neq j$, put $V_{ij} = \Delta_i \Phi\{(e_j - e_i)/h\}$, and set $V_{ii} = 0$. The gradient is

$$c_i = \sum_j V_{ij} - \sum_j V_{ji}, \qquad \nabla f_h(\boldsymbol{\vartheta}) = \widehat{\boldsymbol{W}}^\top \boldsymbol{c}/N. \tag{5.2}$$

The coefficients satisfy $\mathbf{1}_n^\top \boldsymbol{c} = 0$. Event patients anchor rows i, all other patients contribute as j, and the diagonal is excluded from the ordered-pair sum.

For an event-row block $\mathcal{I}$, form the $|\mathcal{I}| \times n$ residual differences, accumulate the row and column contributions to $\boldsymbol{c}$, and sum the associated losses. This evaluates the complete ordered-pair criterion exactly. With $m = \sum_i \Delta_i$ events and block size b, one objective–gradient pass uses $O(nP + mn)$ arithmetic and $O(bn + n + P)$

temporary storage in addition to the stored design. Line-search evaluations contribute further work.

The residual matrix need not be stored either. For any $\boldsymbol{v} = (\boldsymbol{v}_\theta^\top, \boldsymbol{v}_\gamma^\top, \boldsymbol{v}_\beta^\top)^\top$ and $\boldsymbol{c} \in \mathbb{R}^n$, the identities

$$\begin{aligned} \widehat{\boldsymbol{W}}\boldsymbol{v} &= \boldsymbol{Z}_c\boldsymbol{v}_\theta + \widehat{\boldsymbol{F}}(\boldsymbol{v}_\gamma - \widehat{\boldsymbol{B}}^\top\boldsymbol{v}_\beta) + \boldsymbol{X}_c\boldsymbol{v}_\beta, \\ \widehat{\boldsymbol{W}}^\top\boldsymbol{c} &= \begin{pmatrix} \boldsymbol{Z}_c^\top\boldsymbol{c} \\ \widehat{\boldsymbol{F}}^\top\boldsymbol{c} \\ \boldsymbol{X}_c^\top\boldsymbol{c} - \widehat{\boldsymbol{B}}(\widehat{\boldsymbol{F}}^\top\boldsymbol{c}) \end{pmatrix} \end{aligned} \tag{5.3}$$

allow storage of $\boldsymbol{X}_c$ and the low-rank factors in place of a duplicate residual matrix. Euclidean updates remain in the original $(\boldsymbol{a}, \boldsymbol{\beta})$ coordinates.

5.2 Smoothness and a partially penalized proximal step

Let $\mathscr{L}_\Delta$ be the graph Laplacian with edge weights $\Delta_i + \Delta_j$ for $i < j$. Explicitly,

$$\begin{aligned} \mathscr{L}_\Delta &= n\,\mathrm{diag}(\boldsymbol{\Delta}) + m\boldsymbol{I}_n - \boldsymbol{\Delta}\mathbf{1}_n^\top - \mathbf{1}_n\boldsymbol{\Delta}^\top, \\ \mathscr{B}_h &= \frac{\|\widehat{\boldsymbol{W}}^\top\mathscr{L}_\Delta\widehat{\boldsymbol{W}}\|_{\mathrm{op}}}{Nh\sqrt{2\pi}}. \end{aligned} \tag{5.4}$$

Here $\boldsymbol{\Delta} = (\Delta_1, \ldots, \Delta_n)^\top$. Pair grouping and $\phi \le \phi(0)$ imply

$$\begin{aligned} 0 \preceq \nabla^2 f_h(\boldsymbol{\vartheta}) &\preceq \frac{\widehat{\boldsymbol{W}}^\top\mathscr{L}_\Delta\widehat{\boldsymbol{W}}}{Nh\sqrt{2\pi}}, \\ \|\nabla f_h(\boldsymbol{x}) - \nabla f_h(\boldsymbol{y})\|_2 &\le \mathscr{B}_h\|\boldsymbol{x} - \boldsymbol{y}\|_2. \end{aligned} \tag{5.5}$$

This global Lipschitz bound controls numerical step selection. Since $\widehat{\boldsymbol{W}}$ is centered,

$$\begin{aligned} \widehat{\boldsymbol{W}}^\top\mathscr{L}_\Delta\widehat{\boldsymbol{W}} &= n\widehat{\boldsymbol{W}}^\top\,\mathrm{diag}(\boldsymbol{\Delta})\widehat{\boldsymbol{W}} + m\widehat{\boldsymbol{W}}^\top\widehat{\boldsymbol{W}}, \\ \mathscr{B}_h &\le \frac{(n+m)\|\widehat{\boldsymbol{W}}\|_{\mathrm{op}}^2}{Nh\sqrt{2\pi}}. \end{aligned}$$

The Frobenius norm gives a computable upper bound in place of the operator norm. Backtracking also permits step selection without evaluating these norms or forming a full Hessian. The factor $1/h$

governs numerical smoothness, separately from the stochastic scale in Section 4.

For $t > 0$, define the soft-thresholding map by $\mathsf{S}_\tau(z)_j = \mathrm{sign}(z_j)(|z_j| - \tau)_+$. The exact proximal-gradient candidate is

$$\begin{aligned} T_t(\boldsymbol{x}) &= \mathrm{prox}_{tg_\lambda}\{\boldsymbol{x} - t\nabla f_h(\boldsymbol{x})\}, \\ [T_t(\boldsymbol{x})]_a &= \boldsymbol{x}_a - t\nabla_a f_h(\boldsymbol{x}), \qquad [T_t(\boldsymbol{x})]_\beta = \mathsf{S}_{t\lambda}\{\boldsymbol{x}_\beta - t\nabla_\beta f_h(\boldsymbol{x})\}. \end{aligned} \tag{5.6}$$

The proximal map leaves the nuisance block unpenalized. Rescaled residual coordinates use the corresponding weighted threshold from Section 3.

5.3 Backtracking algorithm and deterministic convergence

Choose $t_{\max} > 0$ and $\zeta, \omega \in (0, 1)$. From $\boldsymbol{x} = \boldsymbol{x}^{(r)}$, try $t = t_{\max}$ and repeatedly replace t by ζt until, for $\boldsymbol{y} = T_t(\boldsymbol{x})$ and $\boldsymbol{d} = \boldsymbol{y} - \boldsymbol{x}$,

$$f_h(\boldsymbol{y}) \le f_h(\boldsymbol{x}) + \nabla f_h(\boldsymbol{x})^\top \boldsymbol{d} + \frac{\omega}{2t}\|\boldsymbol{d}\|_2^2. \tag{5.7}$$

A nonnegative remainder bound permits stable acceptance without subtracting nearly equal objective values. Write $d_{ij} = \boldsymbol{w}_{ij}^\top(\boldsymbol{y} - \boldsymbol{x})$ and let m_{ij} be the minimum absolute residual difference along the segment from $r_{ij}(\boldsymbol{x})$ to $r_{ij}(\boldsymbol{y})$. Thus $m_{ij} = 0$ when the segment crosses zero and otherwise equals the smaller endpoint magnitude. Taylor's formula gives

$$\begin{aligned} &f_h(\boldsymbol{y}) - f_h(\boldsymbol{x}) - \nabla f_h(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{x}) \le \mathfrak{B}_h(\boldsymbol{x}, \boldsymbol{y}), \\ &\mathfrak{B}_h(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{2Nh}\sum_{i \ne j} \Delta_i \phi(m_{ij}/h) d_{ij}^2 \\ &\qquad\qquad \le \frac{\mathscr{B}_h}{2}\|\boldsymbol{y} - \boldsymbol{x}\|_2^2. \end{aligned} \tag{5.8}$$

The reference implementation accepts when $\mathfrak{B}_h(\boldsymbol{x}, \boldsymbol{y}) \le \omega\|\boldsymbol{y} - \boldsymbol{x}\|_2^2/(2t)$. This sufficient test passes for $t \le \omega/\mathscr{B}_h$ and can be evaluated in event-row blocks.

Set $t_r = t$ and $\boldsymbol{x}^{(r+1)} = \boldsymbol{y}$. The cap $t_{\max}$ supplies a uniform step-size bound; $\zeta = \omega = 1/2$ is one admissible numerical choice.

Algorithm 1. Fixed-target, partially penalized proximal gradient
Input: $\boldsymbol{Y}, \boldsymbol{\Delta}, \widehat{\boldsymbol{W}}, q$, fixed $h, \lambda > 0$, finite $\boldsymbol{x}^{(0)}$, $t_{\max} > 0$, and $\zeta, \omega \in (0, 1)$.
Preflight: validate dimensions, scales, event coding, PCA rank, and the nuisance-direction diagnostic in Section 5.5. Return a distinct no-event status when the event set is empty.
For $r = 0, 1, \ldots$: compute $f_h(\boldsymbol{x}^{(r)})$ and $\nabla f_h(\boldsymbol{x}^{(r)})$ by blocks; initialize $t = t_{\max}$; form (5.6) and backtrack until (5.7) is certified by (5.8); accept the candidate; recompute its gradient and record the objective, step size, and full optimality residuals. Record the stopping status and full-coordinate residual.
Output: coefficients in the original scale, frozen prediction map, all tuning values, residuals, iteration and evaluation counts, and diagnostic status. Iteration and line-search limits retain separate status codes.

Theorem 5.1 (Convergence for a fixed, attained objective)**.** *Suppose the finite-sample minimizer set $\mathcal{S} = \arg\min Q$ is nonempty. In exact arithmetic, Algorithm 1 accepts a step after finitely many backtracking trials. If $\mathscr{B}_h > 0$, all accepted steps satisfy*

$$t_r \geq t_{\min} := \min\{t_{\max}, \zeta\omega/\mathscr{B}_h\} > 0. \tag{5.9}$$

When $\mathscr{B}_h = 0$, set $t_{\min} = t_{\max}$. The sequence converges to a member of $\mathcal{S}$, and, for each $\boldsymbol{x}^\star \in \mathcal{S}$ and $r \geq 1$,

$$Q(\boldsymbol{x}^{(r+1)}) \leq Q(\boldsymbol{x}^{(r)}) - \frac{2-\omega}{2t_r}\|\boldsymbol{x}^{(r+1)} - \boldsymbol{x}^{(r)}\|_2^2, \tag{5.10}$$

$$0 \leq Q(\boldsymbol{x}^{(r)}) - Q(\boldsymbol{x}^\star) \leq \frac{\|\boldsymbol{x}^{(0)} - \boldsymbol{x}^\star\|_2^2}{2t_{\min} r}. \tag{5.11}$$

No strong convexity or uniqueness of the minimizer is required.

For fixed n, h, λ, Theorem 5.1 gives an $O(r^{-1})$ objective-gap bound for Algorithm 1 in exact arithmetic. Appendix E specializes standard proximal-gradient reasoning (Beck and Teboulle, 2009; Parikh and Boyd, 2014) to the stated line search. Finite numerical stopping uses the residual below.

5.4 Optimality residuals and the finite-tolerance interface

For the Karush–Kuhn–Tucker (KKT) conditions, at a finite candidate $\boldsymbol{x} = (\boldsymbol{a}^\top, \boldsymbol{\beta}^\top)^\top$, let $\boldsymbol{g} = \nabla f_h(\boldsymbol{x})$ and define

$$\begin{aligned} \mathfrak{r}_{\mathrm{KKT}}(\boldsymbol{x}) &= \max\{\|\boldsymbol{g}_a\|_\infty, \max_j \rho_j(\boldsymbol{x})\}, \\ \rho_j(\boldsymbol{x}) &= \begin{cases} |g_{\beta,j} + \lambda \operatorname{sign}(\beta_j)|, & \beta_j \neq 0, \\ (|g_{\beta,j}| - \lambda)_+, & \beta_j = 0. \end{cases} \end{aligned} \tag{5.12}$$

A maximum over an empty block is zero. This equals $\inf\{\|\boldsymbol{s}\|_\infty : \boldsymbol{s} \in \partial Q(\boldsymbol{x})\}$, and is zero if and only if the finite vector $\boldsymbol{x}$ is a global minimizer. The proximal-gradient mapping

$$G_t(\boldsymbol{x}) = \{\boldsymbol{x} - T_t(\boldsymbol{x})\}/t \tag{5.13}$$

provides a second, continuous fixed-point diagnostic, also zero exactly at a minimizer. At an accepted pair $(\boldsymbol{x}, \boldsymbol{y})$, the explicit vector

$$\begin{gathered} \boldsymbol{s}_y = G_t(\boldsymbol{x}) + \nabla f_h(\boldsymbol{y}) - \nabla f_h(\boldsymbol{x}) \in \partial Q(\boldsymbol{y}), \\ \|\boldsymbol{s}_y\|_2 \leq (1 + t\mathscr{B}_h)\|G_t(\boldsymbol{x})\|_2 \end{gathered} \tag{5.14}$$

links the iteration to a subgradient residual. Equation (5.12) uses the stored coefficient's exact zero/nonzero status.

The implementation reports absolute residuals, the declared tolerance, and the normalized residual $\mathfrak{r}_{\mathrm{KKT}}/\max\{\lambda, \|\boldsymbol{g}\|_\infty\}$. Numerical stationarity and nuisance-direction geometry are assessed separately. Supplementary Appendix G specifies the tracked arithmetic-error budget and the distinction between this numerical diagnostic and an objective-gap certificate.

Corollary 5.1 (Finite optimization tolerance on the statistical event)**.** *Under the assumptions and tuning of the ordinary-model conditional theorem in Section 4, let* $k = s + q$, $\boldsymbol{\vartheta}^\dagger = (\boldsymbol{\theta}_0^\top, (\widehat{\boldsymbol{B}}^\top \boldsymbol{\beta}_0)^\top, \boldsymbol{\beta}_0^\top)^\top$, *and* $R = 8\sqrt{k}\lambda/\kappa$. *On the same event used in that theorem, every finite* $\widetilde{\boldsymbol{\vartheta}}$ *satisfying*

$$\mathfrak{r}_{\mathrm{KKT}}(\widetilde{\boldsymbol{\vartheta}}) \leq \lambda/4 \tag{5.15}$$

obeys

$$\begin{gathered} \|\widetilde{\boldsymbol{\vartheta}} - \boldsymbol{\vartheta}^\dagger\|_2 \leq R, \\ \|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_1 \leq 4\sqrt{k}R. \end{gathered} \tag{5.16}$$

Thus the conditional probability is still at least $1 - 2\alpha$*, under the same fixed-tuning hypotheses; no extra event is introduced by this deterministic implication.*

A tilted sublevel argument transfers the Section 4 score and contrast bounds to finite candidates under the same fixed-tuning conditions. The experiments use a tighter numerical threshold of 10^{-7}; outcome-based tuning follows the separate validation procedure of Section 3.6.

5.5 Unpenalized escape directions and finite existence

Define the event-anchored nuisance differences $\boldsymbol{a}_{ij} = \widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_j$ and the flat-direction space

$$\mathcal{N}_A = \{\boldsymbol{v} \in \mathbb{R}^q : \boldsymbol{a}_{ij}^\top \boldsymbol{v} = 0 \text{ whenever } \Delta_i = 1,\ j \neq i\}.$$

A *strict escape direction* is a vector $\boldsymbol{v}$ such that

$$\begin{aligned} \boldsymbol{a}_{ij}^\top \boldsymbol{v} &\leq 0 \quad (\Delta_i = 1,\ j \neq i), \\ \boldsymbol{a}_{ij}^\top \boldsymbol{v} &< 0 \quad \text{for at least one such pair.} \end{aligned} \tag{5.17}$$

Strictness for just one pair suffices because $\ell_h'(r) > 0$ at every finite r when $h > 0$.

Proposition 5.1 (An exact-arithmetic finite-existence criterion)**.** *For finite* $\boldsymbol{Y}, \widehat{\boldsymbol{W}}$*,* $h, \lambda > 0$*, the set* $\arg\min Q$ *is nonempty if and only if no strict escape direction in* (5.17) *exists. If no such direction exists, every nonempty sublevel set is compact after projecting* $\boldsymbol{a}$ *onto* $\mathcal{N}_A^\perp$*. Directions in* $\mathcal{N}_A$ *leave the objective unchanged and need not imply nonexistence. When* $q = 0$*, a finite minimum always exists.*

Proposition 5.1 characterizes finite attainment of the fixed criterion, including flat nuisance directions. The coefficient guarantees in Section 4 additionally impose restricted-design separation.

If $m > 0$, choose an event index i_0 and let $\mathcal{E} = \{i : \Delta_i = 1\}$, $\mathcal{C} = \{j : \Delta_j = 0\}$. The escape test can be expressed with $O(nq)$ coefficients rather than all pair differences: find $\boldsymbol{v} \in \mathbb{R}^q$ satisfying

$$\begin{aligned} (\widehat{\boldsymbol{A}}_i - \widehat{\boldsymbol{A}}_{i_0})^\top \boldsymbol{v} &= 0 \quad (i \in \mathcal{E}), \\ (\widehat{\boldsymbol{A}}_j - \widehat{\boldsymbol{A}}_{i_0})^\top \boldsymbol{v} &\geq 0 \quad (j \in \mathcal{C}), \\ \sum_{j \in \mathcal{C}} (\widehat{\boldsymbol{A}}_j - \widehat{\boldsymbol{A}}_{i_0})^\top \boldsymbol{v} &\geq 1. \end{aligned} \tag{5.18}$$

The last constraint fixes the scale of a positive strict margin. In exact arithmetic, feasibility is equivalent to (5.17). Numerical output includes equality residuals, inequality margins, and an explicit inconclusive status when the margins cannot be resolved.

For $m = 0$, $Q = \lambda\|\boldsymbol{\beta}\|_1$ and every $\boldsymbol{a}$ with $\boldsymbol{\beta} = 0$ is optimal, but the sample provides no event information. For $m = n$, strict escape is impossible because both directions of every comparison are present. Rank deficiency can still leave flat nuisance coordinates. These cases distinguish nonexistence, nonuniqueness, and lack of event information.

5.6 Regularization paths and stable arithmetic

For fixed (K, h), suppose $\boldsymbol{a}^{(0)}$ attains the minimum of $\boldsymbol{a} \mapsto f_h(\boldsymbol{a}, 0)$. The level

$$\lambda_{\text{zero}} = \|\nabla_\beta f_h(\boldsymbol{a}^{(0)}, 0)\|_\infty \tag{5.19}$$

makes $(\boldsymbol{a}^{(0)}, 0)$ a global minimizer for every $\lambda \geq \lambda_{\text{zero}}$ by the full KKT conditions. The threshold is evaluated after checking nuisance-only stationarity. Penalty paths retain strictly positive levels and use the preceding successful fit as a warm start. This construction follows regularized AFT practice (Suder and Molstad, 2022), with the null fit including the unpenalized block.

For a very negative residual difference, evaluating $r\Phi(r/h) + h\phi(r/h)$ naively subtracts nearly equal quantities. A useful stable identity, with $z = |r|/h$, is

$$\ell_h(r) = r_+ + h\phi(z)J(z), \qquad J(z) = \int_0^\infty u \exp(-zu - u^2/2)\,\mathrm{d}u > 0. \tag{5.20}$$

For $z > 0$, an alternating tail expansion is

$$\begin{aligned} J(z) &= \sum_{j=0}^{M-1} \frac{(-1)^j (2j+1)!!}{z^{2j+2}} + \mathcal{E}_M(z), \\ |\mathcal{E}_M(z)| &\le \frac{(2M+1)!!}{z^{2M+2}}. \end{aligned} \tag{5.21}$$

Here M is the expansion length. The implementation combines normal-tail routines, log-scale evaluation, and the explicit remainder bound. Numerical metadata track tail approximation, unresolved arithmetic, and the line-search budget; details appear in Supplementary Appendix G.

Every candidate retains its declared objective, complete-coordinate stopping rule, and frozen prediction map. Output states distinguish stationarity to tolerance, iteration limits, numerical or line-search failures, nuisance escape, flat directions, and event-free samples.

5.7 Accelerated proximal iterations

The numerical studies use the non-restarted backtracking fast iterative shrinkage-thresholding algorithm (FISTA) (Beck and Teboulle, 2009). Let $\boldsymbol{x}^0 = \boldsymbol{y}^0$ and $\tau_0 = 1$. At iteration r, the smooth gradient is evaluated at the extrapolated point $\boldsymbol{y}^r$. Starting from the previous accepted inverse step L_{r-1}, increase L_r geometrically until the partially penalized candidate

$$\begin{aligned} \boldsymbol{x}_a^{r+1} &= \boldsymbol{y}_a^r - L_r^{-1} \nabla_a f_h(\boldsymbol{y}^r), \\ \boldsymbol{x}_\beta^{r+1} &= \mathsf{S}_{\lambda/L_r}\{\boldsymbol{y}_\beta^r - L_r^{-1} \nabla_\beta f_h(\boldsymbol{y}^r)\} \end{aligned} \tag{5.22}$$

satisfies the full-segment sufficient check

$$\mathfrak{B}_h(\boldsymbol{y}^r, \boldsymbol{x}^{r+1}) \le \frac{\omega L_r}{2} \|\boldsymbol{x}^{r+1} - \boldsymbol{y}^r\|_2^2. \tag{5.23}$$

Here $\mathfrak{B}_h$ is defined in (5.8). The implemented values are $\omega = 1/2$, an initial inverse step of 1, and a doubling factor of 2. The accepted L_r is carried forward and does not decrease. After acceptance,

$$\begin{aligned} \tau_{r+1} &= \frac{1+\sqrt{1+4\tau_r^2}}{2}, \\ \boldsymbol{y}^{r+1} &= \boldsymbol{x}^{r+1} + \frac{\tau_r - 1}{\tau_{r+1}}(\boldsymbol{x}^{r+1} - \boldsymbol{x}^r). \end{aligned} \tag{5.24}$$

The full gradient and KKT residual are recomputed at the returned point $\boldsymbol{x}^{r+1}$. Acceptance controls the quadratic majorant at the extrapolated point $\boldsymbol{y}^r$, so successive accepted objectives can increase. Theorem 5.1 applies to the reference Algorithm 1. The accelerated implementation uses the same objective and full-coordinate finite-tolerance interface to the statistical analysis.

5.8 Qualified active-face Newton refinement

An approximately identified sparse face can be refined using its local curvature. This uses standard second-order ideas for composite optimization (Lee et al., 2014). At a finite input point, define

$$\mathcal{A} = \{1, \ldots, q\} \cup \{j > q : \vartheta_j \neq 0\}, \qquad Z = \{j > q : \vartheta_j = 0\}.$$

Retain this set and the signs of the penalized active coordinates during the local call. Put $v_j = 0$ for the unpenalized coordinates and $v_j = \operatorname{sign}(\vartheta_j)$ for the nonzero penalized coordinates. With $\boldsymbol{g} = \nabla f_h(\boldsymbol{\vartheta})$ and $\boldsymbol{H} = \nabla^2 f_h(\boldsymbol{\vartheta})$, solve

$$\begin{aligned} \boldsymbol{H}_{\mathcal{A}\mathcal{A}}\boldsymbol{d}_{\mathcal{A}} &= -(\boldsymbol{g}_{\mathcal{A}} + \lambda \boldsymbol{v}_{\mathcal{A}}), \\ \boldsymbol{d}_Z &= 0. \end{aligned} \tag{5.25}$$

The numerically positive-definite active system is solved directly by Cholesky factorization.

Let $\widehat{s}_{\mathcal{A}} = \widehat{\boldsymbol{g}}_{\mathcal{A}} + \lambda \boldsymbol{v}_{\mathcal{A}}$ and let E denote the tracked gradient error budget. For a trial $t > 0$ inside the original sign face, use the sufficient Armijo check

$$\begin{aligned} t\widehat{s}_{\mathcal{A}}^{\top}\boldsymbol{d}_{\mathcal{A}} + \mathfrak{B}_h(\boldsymbol{\vartheta}, \boldsymbol{\vartheta} + t\boldsymbol{d}) + tE\|\boldsymbol{d}\|_1 &\leq c_A t\widehat{s}_{\mathcal{A}}^{\top}\boldsymbol{d}_{\mathcal{A}}, \\ c_A &= 10^{-4}. \end{aligned} \tag{5.26}$$

Trial steps are halved as necessary. A 0.99 fraction-to-boundary rule retains the input signs, and every accepted candidate is checked on all original coordinates, including the numerical condition $|\widehat{g}_j| + E \leq \lambda$ for every $j \in Z$. Here E is the candidate's tracked gradient error budget. The local call stops if the fixed face, numerical curvature, or line-search requirements cease to hold. Passing the complete KKT test is required for a successful output.

5.9 Common numerical policy for training comparisons

Both estimators use the same sequence of numerical operations in the training-validation study. Each candidate first receives up to 30,000 accepted FISTA updates at full-coordinate tolerance 10^{-7}. A successful return is used directly. A return at the update limit qualifies for at most one active-face refinement if its last five distinct saved accepted checkpoints have identical exact $\boldsymbol{\beta}$ support and signs, the gradient budget is at most 10^{-8}, and all zero-coordinate margins remain strictly positive after allowing for that budget. The refinement allows at most ten accepted Newton updates. Otherwise the candidate is recorded as numerically ineligible under the specified budget.

The final parameter vector defines all coefficient blocks, the deployment slope (3.17), fitted scores, and full KKT residual. After refinement, these quantities are reconstructed from the new coefficients and unchanged design. The preceding successful candidate supplies the next warm start. This policy and the selection rule jointly define the procedure evaluated in Section 6.4.

6 Computational illustrations and training selection

The numerical study addresses two questions: how factor coordinates relate to computational geometry, and how training selection allocates fitted effects between common and sparse residual components. Six synthetic samples, one per scenario, provide fixed inputs for these illustrations. Curvature and stationarity diagnostics address the first question; three-fold validation and final-model summaries address the second.

6.1 Synthetic design and comparison protocol

Each cohort has $n = 120$ training patients, $p = 300$ molecular covariates, and $d = 2$ clinical covariates. Independently generate the coordinates of $\boldsymbol{f}_i \in \mathbb{R}^2$, $\boldsymbol{v}_i \in \mathbb{R}^p$, and $\boldsymbol{Z}_i \in \mathbb{R}^2$ from $\mathrm{Unif}[-\sqrt{3}, \sqrt{3}]$. For $j = 1, \ldots, p$, let

$$\begin{aligned} \boldsymbol{B}_{j\cdot} &= \sqrt{\rho}\left(\cos\frac{2\pi(j-1)}{p},\ \sin\frac{2\pi(j-1)}{p}\right), \\ \boldsymbol{u}_i &= \sqrt{1-\rho}\,\boldsymbol{v}_i, \qquad \boldsymbol{X}_i = \boldsymbol{B}\boldsymbol{f}_i + \boldsymbol{u}_i. \end{aligned} \tag{6.1}$$

The ordinary-model signal is $\eta_i = \boldsymbol{Z}_i^\top \boldsymbol{\theta}_0 + \boldsymbol{X}_i^\top \boldsymbol{\beta}_0$, with $\boldsymbol{\theta}_0 = (0.35, -0.25)^\top$ and nonzero $\boldsymbol{\beta}_0$ entries $(0.6, -0.6, 0.5, -0.5, 0.4)$ at indices $(1, 61, 121, 181, 241)$. All remaining entries are zero. The error is standard normal, $a_3 t_3$, or $a_C C$ with C standard Cauchy, where

$$a_3 = \Phi^{-1}(0.75)/F_{t_3}^{-1}(0.75), \qquad a_C = \Phi^{-1}(0.75).$$

Thus the error interquartile ranges (IQRs) agree. Independently draw $V_i \sim N(0, 1)$ and set

$$\begin{aligned} \log T_i &= \eta_i + \varepsilon_i, \\ \log C_i &= \eta_i + c + V_i, \\ \int F_\varepsilon(c+v)\phi(v)\,\mathrm{d}v &= 1 - \pi_C. \end{aligned} \tag{6.2}$$

Equation (6.2) calibrates the population censoring proportion π_C. Realized event indicators follow the generated event and censoring times, retaining sample variation in censoring. Computation remains on the log-time scale.

The factor-adjusted estimator (FA) uses $K = 2$ and the unadjusted estimator (UA) uses $K = 0$; the loss, molecular penalty, and bandwidth $h = 0.2$ agree. S1 is the prespecified over-adjustment control, with $\rho = 0$. At the fixed coefficients, signal variance is approximately 1.565, 1.3254, and 0.8462 for $\rho = 0, 0.25, 0.75$. The scenarios therefore vary dependence at fixed coefficients, with the corresponding change in signal variance.

Table 1 Synthetic-design scenarios. The error scales match interquartile ranges

Scenario	ρ	Target censoring	Error distribution
S1	0	30%	Standard normal
S2	0.25	30%	Standard normal
S3	0.75	30%	Standard normal
S4	0.75	30%	IQR-scaled t_3
S5	0.75	60%	IQR-scaled t_3
S6	0.75	30%	IQR-scaled Cauchy

6.2 Active curvature and work to common accuracy

Both estimators use non-restarted FISTA at $\lambda = 0.01$, initialized from their respective $\lambda = 0.025$ fits on S3, S4, and S6. Each method retains its original parameter coordinates and training design. The common stopping threshold is 10^{-7} for the complete KKT residual plus the tracked numerical budget.

Factor adjustment reaches the common accuracy levels with fewer gradient evaluations in these three samples (Figure 1). At 10^{-6}, first-crossing counts for FA and UA are 2,549 and 33,195 in S3, and 3,021 and 34,693 in S4. S6 FA reaches this level at 4,843 evaluations; S6 UA remains above it at its endpoint. At the prescribed 10^{-7} stopping level, FA takes 3,619, 3,271, and 4,507 accepted updates. UA reaches its 30,000-update limit with residuals 4.269×10^{-7}, 3.033×10^{-7}, and 2.686×10^{-6}. The accepted-update view is provided in Supplementary Appendix F.

For each saved endpoint, form the active set from all unpenalized coordinates and the exactly nonzero molecular coefficients. The local loss Hessian is

$$\boldsymbol{H}(\boldsymbol{\vartheta}) = \frac{1}{Nh} \sum_{i \neq j} \Delta_i \phi\{r_{ij}(\boldsymbol{\vartheta})/h\} \, \boldsymbol{w}_{ij} \boldsymbol{w}_{ij}^{\top}. \tag{6.3}$$

We analyze $\boldsymbol{H}_{\mathcal{A}\mathcal{A}}$ using the numerical rank threshold

$$100 \epsilon_{\text{mach}} |\mathcal{A}| \max_j |\lambda_j(\boldsymbol{H}_{\mathcal{A}\mathcal{A}})|.$$

All six active Hessians were numerically full rank under this rule.

The spectral differences are driven primarily by the upper end: FA reduces the maximum active curvature from 89–116 to 2.89–3.70, with smaller active spectral ratios in all three samples (Table 2).

Table 2 Active loss-Hessian spectra at the FISTA endpoints. Ratios use the smallest numerical positive eigenvalue

Scenario	Method	Dimension	Minimum	Maximum	Ratio
S3	FA	103	9.055×10^{-4}	2.893	3.195×10^{3}
S3	UA	101	1.055×10^{-3}	89.074	8.445×10^{4}
S4	FA	101	2.188×10^{-3}	2.888	1.320×10^{3}
S4	UA	100	1.508×10^{-3}	94.353	6.256×10^{4}
S6	FA	102	4.942×10^{-4}	3.704	7.494×10^{3}
S6	UA	104	2.463×10^{-4}	115.830	4.702×10^{5}

Minimum curvature varies by scenario and is slightly lower for FA in S3. The corresponding final scalar steps are 1/4 for FA and 1/256 for UA. UA's inverse step remains $L = 256$ throughout its last 5,000 updates, with no further backtracking. The lower upper-curvature scale is therefore consistent with the larger scalar steps used by FA.

The largest endpoint residual lies in a nonzero molecular coefficient for all six fits. Exactly zero coordinates satisfy the KKT inequalities at the endpoints and selected late checkpoints. Across UA updates 26,000, 27,000, 28,000, 29,000, and 30,000, the saved supports and signs agree while the dominant residual coordinate changes. Clinical-block residuals also exceed 10^{-7} for S3 and S6 UA, confirming the need for full-coordinate accuracy.

Weak-curvature directions explain where the remaining stationarity error lies. The ten smallest-eigenvalue directions contain 79.1%, 80.3%, and 52.8% of the squared active residual for S3, S4, and S6 UA. FA residuals also concentrate in weak directions. For UA, this concentration combines with a wider active spectrum and a fixed scalar step, producing small late objective ranges alongside oscillating KKT residuals. These comparisons use each method's own endpoint, active set, and optimization domain.

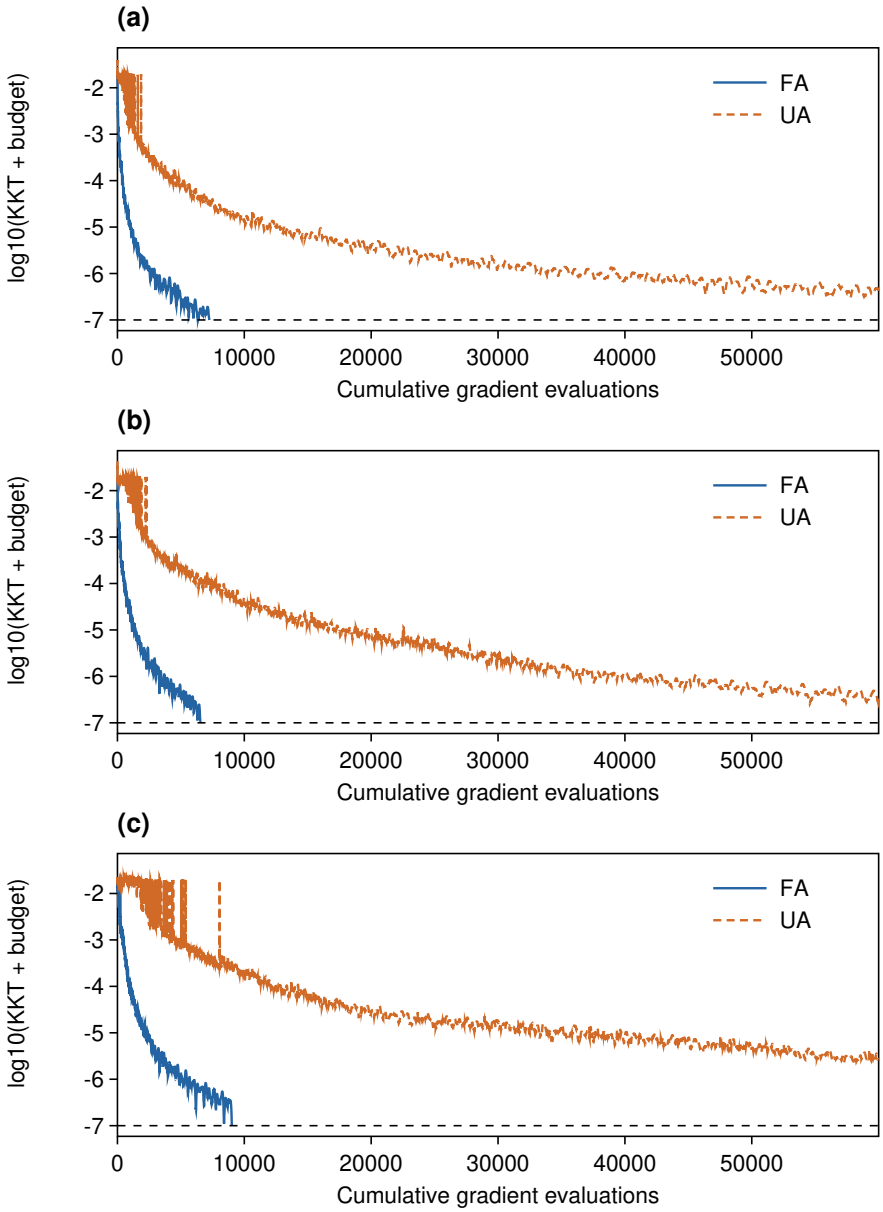


Fig. 1 Computational work to a common stationarity tolerance. The full original-coordinate KKT residual plus its tracked numerical budget is plotted against cumulative gradient evaluations for S3, S4, and S6 in panels (a), (b), and (c). FA is solid and UA is dashed. Both methods use non-restarted FISTA, $h = 0.2$, $\lambda = 0.01$, and their respective $\lambda = 0.025$ starts. The horizontal line is 10^{-7}. Curves end at their FISTA stopping points, before the local refinement in Section 6.3

6.3 Active-face refinement resolves late-stage residuals

One active-face Newton step brings each UA endpoint below 10^{-7} while preserving its design, support, signs, $h = 0.2$, and $\lambda = 0.01$ (Table 3). All three FA endpoints meet the same accuracy threshold before refinement and return unchanged.

Table 3 Full original-coordinate KKT residuals for one-step refinement of the UA fits

Scenario	Before	After	Newton updates
S3	4.269×10^{-7}	4.864×10^{-10}	1
S4	3.033×10^{-7}	1.780×10^{-10}	1
S6	2.686×10^{-6}	6.997×10^{-9}	1

All three step lengths were one, with no backtracking halvings. The full steps preserved every active molecular sign and left all original zero coefficients at zero. The new minimum inactive-coordinate margins were approximately 7.75×10^{-5}, 2.76×10^{-6}, and 9.99×10^{-6}. The objective decreases were approximately 2.58×10^{-10}, 1.02×10^{-10}, and 4.94×10^{-9}. Thus substantial reductions in full stationarity residual were obtained without changing the support or statistical objective.

The full steps provide a direct computational counterpart to the spectral diagnosis: local direction-specific scaling completes stationarity on each existing active face. The training-selection procedure below offers the same refinement rule to both estimators.

6.4 Training selection and factor-mediated prediction

Each of the six 120-patient training cohorts was split once into three event-stratified folds of 40 patients; both methods used the same labels. For every fold and method, centering and PCA were learned on the 80 training patients, and validation predictions used only the resulting frozen transformation. The candidate grid was the fixed absolute set $\{0.15, 0.06, 0.025, 0.01\}$ at $h = 0.2$. The validation score was (3.18), with the zero-prediction contrast computed within each ordered pair and aggregated over the three within-fold pair sets.

Each method selected among candidates valid in all three folds. One invalid fold made the aggregate unavailable. Candidates within $10^{-10} + 10^{-10}\max(1, |\mathrm{CV}_{\min}|)$ of the best valid score were tied, with larger λ preferred. After selection, each full-training refit used freshly estimated preprocessing and a new nuisance initialization.

The procedure produced 143 successful fold candidates out of 144 attempted positions, giving 47 eligible three-fold aggregates out of 48. All 12 selected full-training refits met the full-coordinate KKT-plus-budget threshold of 10^{-7}. Table 4 reports the selections and refit residuals. Figure 2 shows all four penalty positions for each method, marking the selected points and the invalid position.

Table 4 Training-only penalty selections and full-training refit residuals. A star denotes the upper grid boundary

	FA			UA		
Scenario	Eligible	λ	KKT + budget	Eligible	λ	KKT + budget
S1	4/4	0.06	9.825×10^{-8}	4/4	0.06	7.786×10^{-8}
S2	4/4	0.06	9.600×10^{-8}	4/4	0.06	8.981×10^{-8}
S3	4/4	0.06	9.886×10^{-8}	4/4	0.06	9.950×10^{-8}
S4	4/4	0.15*	9.323×10^{-10}	3/4	0.06	9.870×10^{-8}
S5	4/4	0.06	9.503×10^{-10}	4/4	0.025	9.925×10^{-8}
S6	4/4	0.15*	7.446×10^{-10}	4/4	0.15*	9.298×10^{-8}

Both methods select $\lambda = 0.06$ in S1–S3 and 0.15 in S6, with different choices in S4 and S5. Three selections are at the upper grid boundary. S4 fold 1 UA at $\lambda = 0.01$ reaches the FISTA budget without satisfying the support-and-sign refinement gate, so that candidate has no aggregate score. At the selected penalties, FA has a lower validation score in S2, and UA has a lower score in S1 and S3–S6 (Table 5). The scores describe the stated training partitions.

The selected models reveal a structural distinction between residual sparsity and molecular prediction. In S4–S6, FA retains factor-mediated prediction with $\widehat{\boldsymbol{\beta}} = 0$; its deployment slope has 300 exactly nonzero entries in every scenario. UA's deployment slope equals its molecular coefficient vector and shares its nonzero count (Table 5). These quantities summarize different roles in the fitted model: sparse residual effects and total molecular information in the frozen score.

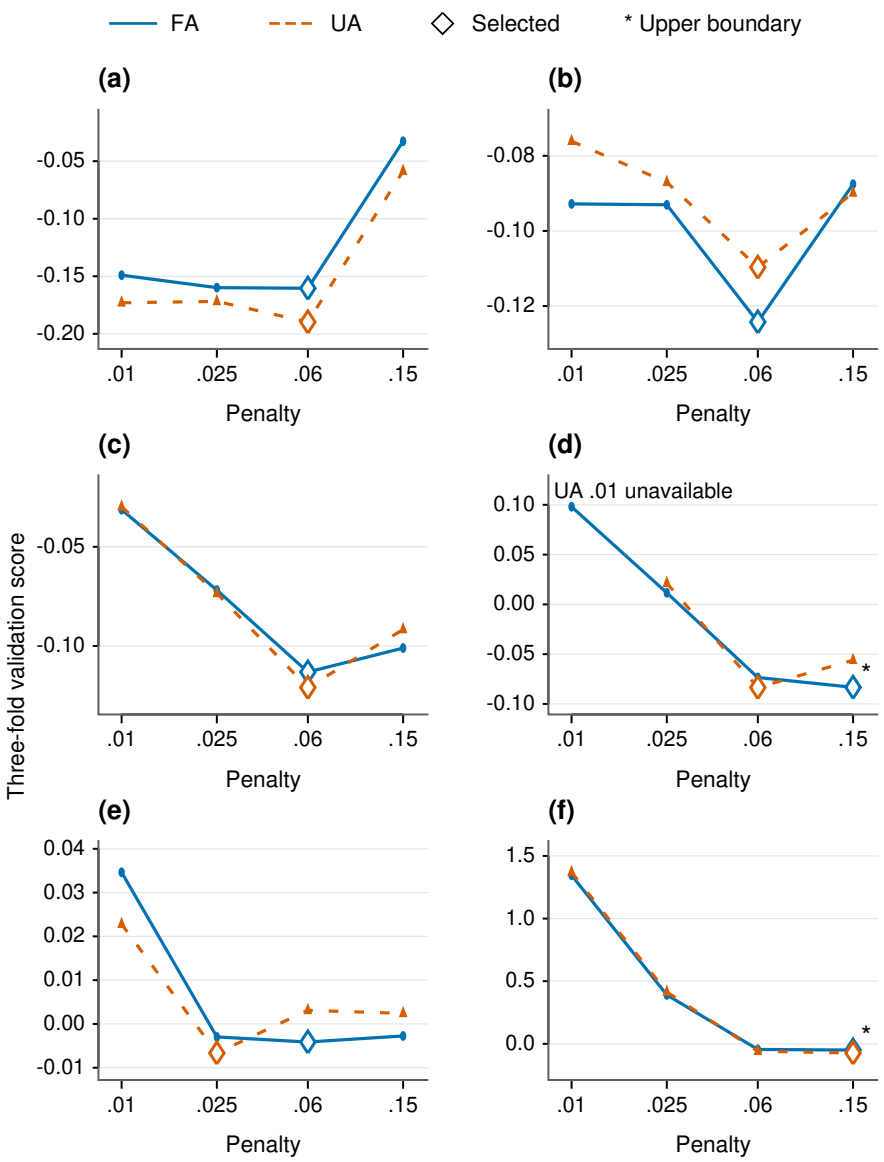


Fig. 2 Training-only three-fold validation paths on S1–S6 in panels (a)–(f). FA uses solid lines and circles; UA uses dashed lines and triangles. Each score is the aggregate within-fold Gehan loss contrast against zero prediction; lower values are preferred. Diamonds mark selected penalties and asterisks mark the upper boundary of the original fixed grid. The S4 UA candidate at $\lambda = 0.01$ has no aggregate score because one fold failed its numerical eligibility rule. Each panel uses one three-fold partition and its own vertical scale

Table 5 Selected full-training models. $\|\hat{\boldsymbol{\beta}}\|_0$ counts exact nonzero penalized coefficients; $\|\hat{\boldsymbol{b}}_{\mathrm{pred}}\|_0$ counts exact nonzero frozen prediction slopes. CV is the three-fold score at the selected penalty; smaller is better. FISTA and Newton columns count accepted updates in the full-training refit

Scenario	Method	CV	$\|\hat{\boldsymbol{\beta}}\|_0$	$\|\hat{\boldsymbol{b}}_{\mathrm{pred}}\|_0$	FISTA	Newton
S1	FA	-0.160420	58	300	449	0
S1	UA	-0.189698	55	55	314	0
S2	FA	-0.124238	36	300	209	0
S2	UA	-0.109671	38	38	1439	0
S3	FA	-0.113124	6	300	145	0
S3	UA	-0.120923	11	11	9178	0
S4	FA	-0.083292	0	300	0	0
S4	UA	-0.083487	7	7	7184	0
S5	FA	-0.004137	0	300	0	0
S5	UA	-0.006646	23	23	12054	0
S6	FA	-0.049824	0	300	0	0
S6	UA	-0.073506	2	2	1755	0

Nine UA fold candidates reach the prescribed accuracy after one qualified Newton update each. All twelve selected full-training fits finish without refinement. The S4–S6 FA refits satisfy the stopping test at their new nuisance initializations and take zero FISTA updates. Table 5 reports refit work; Supplementary Appendix F includes initialization and complete work accounting.

7 Discussion

The finite-sample analysis connects same-sample factor learning to estimation of the original sparse AFT slope. The reference representation preserves the ordinary-model predictor exactly, placing the effect of factor estimation in the design geometry. Conditional score centering, column-scale empirical-process control, and local event information then yield coefficient bounds and finite attainment under explicit assumptions. Additional factor effects enter through a separate approximation term relative to the specified latent representation.

This separation is also useful computationally. In the fixed strong-dependence illustrations, FA has lower maximum active curvature and smaller active spectral ratios, together with larger scalar steps

and less work to common accuracy. Late UA residuals concentrate in weak-curvature directions on the selected stable supports. A single active-face Newton correction supplies the direction-specific scaling needed to complete each of those fits. The numerical policy combines proximal search with qualified local refinement, using the same full-coordinate accuracy criterion for both estimators.

Training selection gives complementary information about the representation. Some FA fits retain a dense factor-mediated molecular score with a zero residual coefficient block. Thus $\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{b}}_{\mathrm{pred}}$ answer different questions: the former describes residual associations, while the latter records the molecular slope deployed for prediction. The six training partitions exhibit different selected-score orderings across methods, making both coefficient recovery and independent prediction relevant to evaluation of the complete selection procedure.

The theoretical bounds concern fixed or covariate-measurable tuning, and the numerical experiments describe fixed-sample geometry and training selection. Independent repeated samples can assess the risk of the outcome-selected procedure under prespecified grids and numerical rules. A biomedical application would additionally require a common endpoint, an explicit observation scheme, and evaluation of the frozen prediction map on an external cohort. These evaluations extend the present theory and computational illustrations to statistical performance and applied use.

The resulting framework links three objects: the original sparse AFT target, covariate geometry learned from the sample, and the numerical accuracy of the fitted estimator. Exact reference coordinates preserve the target, finite-sample analysis accounts for censoring and factor estimation, and local curvature completes the computational connection. Together, they provide a coherent basis for sparse censored regression with estimated factor structure.

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Ethics approval The work develops statistical theory and uses synthetic data. No human participants, animals, or identifiable personal data were studied.

Consent to participate and consent for publication Not applicable: the study contains no human-participant data or identifiable personal information.

Data and code availability Data-generating mechanisms and numerical procedures are described in the article and Online Resource 1. [Author confirmation required: supply an accessible repository and persistent identifier, or authorize an accurate availability-on-request statement for the synthetic data and implementation.]

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Supplementary information Online Resource 1 contains proofs, numerical evidence descriptions, implementation details, and the supplementary accepted-update trajectory figure.

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